

Sid Mathur's Thematic ETF Screener – Daily Report

2026-05-12

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Action Center**Run date:** 2026-05-12 **Universe:** 668 ETFs / 26 clusters**Full PC1:** LONG since 2026-04-13 (D+20, 4.0%) **Clusters long:** 14/26 **ETFs long:** 448/668 (67%)**Today's events****BUY (21)**

level	id	name	category	cluster_signal	spread_buy	prob_buy	nss
etf	SIVR	abrdrn Physical Silver Shares ...	Silver	FLAT	2.057	0.95	0.796
etf	SLV	iShares Silver Trust	Silver	FLAT	1.939	0.95	0.794
etf	KWEB	KraneShares CSI China Interne...	China Equity	FLAT	0.209	0.94	0.656
etf	LTPZ	Pimco 15 Year U.S. TIPS Index...	Inflation-Protected...	LONG	0.047	0.87	0.292
etf	PPLT	abrdrn Physical Platinum Share...	Platinum	FLAT	0.746	0.66	0.155
etf	SILJ	Amplify Junior Silver Miners ...	Materials	FLAT	0.041	0.66	0.141
etf	BAB	Invesco Taxable Municipal Bon...	Taxable Municipal B...	LONG	0.004	0.74	0.098
etf	SIL	Global X Silver Miners ETF	Materials	FLAT	0.735	0.68	0.096
etf	SUB	iShares Short-Term National M...	Municipal Bond	FLAT	0.019	0.90	0.003
etf	VMBS	Vanguard Mortgage-Backed Secu...	Mortgage-Backed Fix...	LONG	0.055	0.78	-0.083
etf	BOND	PIMCO Active Bond Exchange-Tr...	Active Fixed Income	LONG	0.090	0.81	-0.105
etf	MBB	iShares MBS ETF	Mortgage-Backed Fix...	LONG	0.078	0.70	-0.169
cluster	cluster_5	Mixed: BND, AGG, SCHZ	Mixed		0.035	0.70	-0.177
etf	GVI	iShares Intermediate Governme...	Fixed Income	LONG	0.034	0.70	-0.224
etf	AGG	iShares Core U.S. Aggregate B...	Aggregate Fixed Inc...	LONG	0.028	0.65	-0.232
etf	BND	Vanguard Total Bond Market ETF	Aggregate Fixed Inc...	LONG	0.017	0.64	-0.256
etf	SPAB	State Street SPDR Portfolio A...	Aggregate Fixed Inc...	LONG	0.003	0.59	-0.299
etf	SCHZ	Schwab US Aggregate Bond ETF	Aggregate Fixed Inc...	LONG	0.002	0.58	-0.343
etf	NANR	State Street SPDR S&P North A...	Materials	FLAT	0.328	0.35	-0.351
etf	SPMB	State Street SPDR Portfolio M...	Mortgage-Backed Fix...	LONG	0.005	0.53	-0.358
etf	AGZ	iShares Agency Bond ETF	Agency Fixed Income	LONG	0.013	0.45	-0.526

SELL (4)

level	id	name	category	cluster_signal	spread_sell	prob_sell	nss
etf	LVHD	Franklin U.S. Low Volatility ...	U.S. Low Volatility...	LONG	-0.260	0.99	-0.952
etf	FDL	First Trust Morningstar ETF	U.S. Dividend Equity	LONG	-0.277	0.91	-0.883
etf	PKW	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	LONG	-0.691	0.97	0.033
etf	IPAY	Amplify Digital Payments ETF	Financials	FLAT	-0.544	0.82	0.165

Imminent (cluster level, $p > 50\%$)**Imminent BUY (top 5)**

id	name	category	spread_buy	prob_buy	nss
cluster_23	Cash / Ultra-Short Fixed Inco...	Cash / Ultra-Short Fi...	0.393	1	1.000
cluster_22	Mixed: SRLN, BKLN, FTSL	Mixed	1.305	1	1.000
cluster_3	Technology (34/63)	Technology	17.497	1	1.000
cluster_8	Emerging Markets Equity (19/2...	Emerging Markets Equi...	11.512	1	1.000
cluster_4	Mixed: IWB, ITOT, SCHB	Mixed	8.861	1	0.987

Imminent SELL (top 5)

id	name	category	spread_sell	prob_sell	nss
cluster_26	Currency (1/2)	Currency	-1.324	1.00	-1.000
cluster_17	Consumer Staples (6/11)	Consumer Staples	-0.034	0.74	-0.232
cluster_15	Mixed: DBP, GLTR, GDXJ	Mixed	0.401	0.61	-0.193
cluster_11	Healthcare (16/17)	Healthcare	-0.362	0.87	0.021
cluster_18	Mixed: ILF, EWZ, ECH	Mixed	-0.544	0.91	0.072

Full Sample PC1



Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages (last 12 months).

Current state: LONG since 2026-04-13 (20 days, 4.0%)

5dma: 140.590 20dma: 138.541 50dma: 134.180

Spread vs 50dma: +6.410 Spread vs 20dma: +2.050

P(buy crossover within 5 days): 100.0% P(sell): 4.3%

Top 10 PC1 weights

	ticker	weight	name	category
102	ACWI	0.0021	iShares MSCI ACWI ETF	Global Equity
113	AOA	0.0021	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation
133	SPGM	0.0021	State Street SPDR Portfolio MSCI Global...	Global Equity
137	VT	0.0021	Vanguard Total World Stock Index ETF	Global Equity
46	SPY	0.0020	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity
116	AOR	0.0020	iShares Core 60/40 Balanced Allocation ...	Asset Allocation
120	URTH	0.0020	iShares MSCI World ETF	Global Equity

122	CRBN	0.0020	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity
180	RSP	0.0020	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity
264	IMCG	0.0020	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth

Position Sheet

Replicate each currently-long cluster via its 3-5 member basket. OLS weights sum to ~1 per cluster for unit-variance replication; scale each cluster's basket to your target cluster allocation. sticky=yes means members are unchanged from the prior run (only weights drift); sticky=no means a fresh greedy selection.

Active positions (long clusters)

ticker	cluster	theme	cluster_nss	days_long	cum_return	basket_weight	basket_cor	sticky	etf_signal
IEMG	cluster_8	Emerging Markets Equity (19...	1.000	21	9.9%	0.776	0.99	no	LONG
SRLN	cluster_22	Mixed: SRLN, BKLN, FTSL	1.000	24	1.2%	0.311	0.89	no	LONG
FLOT	cluster_22	Mixed: SRLN, BKLN, FTSL	1.000	24	1.2%	0.292	0.89	no	FLAT
BKLN	cluster_22	Mixed: SRLN, BKLN, FTSL	1.000	24	1.2%	0.209	0.89	no	LONG
VWO	cluster_8	Emerging Markets Equity (19...	1.000	21	9.9%	0.178	0.99	no	LONG
FLRN	cluster_22	Mixed: SRLN, BKLN, FTSL	1.000	24	1.2%	0.127	0.89	no	FLAT
FLTR	cluster_22	Mixed: SRLN, BKLN, FTSL	1.000	24	1.2%	0.034	0.89	no	FLAT
EEM	cluster_8	Emerging Markets Equity (19...	1.000	21	9.9%	-0.012	0.99	no	LONG
VOO	cluster_4	Mixed: IWB, ITOT, SCHB	0.987	20	5.9%	0.804	0.99	no	LONG
IVV	cluster_4	Mixed: IWB, ITOT, SCHB	0.987	20	5.9%	0.309	0.99	no	LONG
SPY	cluster_4	Mixed: IWB, ITOT, SCHB	0.987	20	5.9%	-0.154	0.99	no	LONG
VTV	cluster_6	Mixed: IWD, SCHV, VONV	0.919	20	2.6%	0.546	1.00	no	LONG
VO	cluster_6	Mixed: IWD, SCHV, VONV	0.919	20	2.6%	0.278	1.00	no	LONG
VIG	cluster_6	Mixed: IWD, SCHV, VONV	0.919	20	2.6%	0.158	1.00	no	LONG
DBC	cluster_21	Broad Commodities (6/9)	0.918	84	47.8%	0.662	0.98	no	LONG
PDBC	cluster_21	Broad Commodities (6/9)	0.918	84	47.8%	0.535	0.98	no	LONG
FTGC	cluster_21	Broad Commodities (6/9)	0.918	84	47.8%	-0.029	0.98	no	FLAT
VB	cluster_2	Mixed: VB, SCHM, IJH	0.849	20	3.7%	0.582	0.99	no	LONG
IJH	cluster_2	Mixed: VB, SCHM, IJH	0.849	20	3.7%	0.331	0.99	no	LONG
IWM	cluster_2	Mixed: VB, SCHM, IJH	0.849	20	3.7%	0.067	0.99	no	LONG
SCHH	cluster_14	Real Estate (14/14)	0.816	21	4.2%	0.537	1.00	no	LONG
VNQ	cluster_14	Real Estate (14/14)	0.816	21	4.2%	0.515	1.00	no	LONG
XLRE	cluster_14	Real Estate (14/14)	0.816	21	4.2%	-0.075	1.00	no	LONG
SCHP	cluster_19	Inflation-Protected Fixed I...	0.787	22	0.8%	0.748	0.99	no	LONG
DVY	cluster_10	U.S. Dividend Equity (11/20)	0.787	14	0.6%	0.615	0.98	no	LONG
HDV	cluster_10	U.S. Dividend Equity (11/20)	0.787	14	0.6%	0.236	0.98	no	LONG
STIP	cluster_19	Inflation-Protected Fixed I...	0.787	22	0.8%	0.168	0.99	no	FLAT
SCHD	cluster_10	U.S. Dividend Equity (11/20)	0.787	14	0.6%	0.131	0.98	no	LONG
VTIP	cluster_19	Inflation-Protected Fixed I...	0.787	22	0.8%	0.127	0.99	no	FLAT
HYG	cluster_9	High Yield Bond (10) + Pref...	0.489	20	0.7%	0.476	0.99	no	LONG
PFF	cluster_9	High Yield Bond (10) + Pref...	0.489	20	0.7%	0.260	0.99	no	LONG
EMB	cluster_9	High Yield Bond (10) + Pref...	0.489	20	0.7%	0.157	0.99	no	FLAT
VBR	cluster_7	Mixed: VBR, MDYV, IVOV	0.380	21	2.9%	0.430	1.00	no	LONG
IJR	cluster_7	Mixed: VBR, MDYV, IVOV	0.380	21	2.9%	0.363	1.00	no	LONG

XLF	cluster_7	Mixed: VBR, MDYV, IVOV	0.380	21	2.9%	0.219	1.00	no	FLAT
VHT	cluster_11	Healthcare (16/17)	0.021	3	-2.0%	1.989	0.99	no	FLAT
XBI	cluster_11	Healthcare (16/17)	0.021	3	-2.0%	0.128	0.99	no	FLAT
XLV	cluster_11	Healthcare (16/17)	0.021	3	-2.0%	-1.104	0.99	no	FLAT
BND	cluster_5	Mixed: BND, AGG, SCHZ	-0.177	0	0.0%	0.629	1.00	no	LONG
AGG	cluster_5	Mixed: BND, AGG, SCHZ	-0.177	0	0.0%	0.354	1.00	no	LONG
BNDX	cluster_5	Mixed: BND, AGG, SCHZ	-0.177	0	0.0%	0.033	1.00	no	LONG
VPU	cluster_17	Consumer Staples (6/11)	-0.232	6	-2.3%	0.703	0.99	no	FLAT
XLP	cluster_17	Consumer Staples (6/11)	-0.232	6	-2.3%	0.584	0.99	no	LONG
XLU	cluster_17	Consumer Staples (6/11)	-0.232	6	-2.3%	-0.324	0.99	no	FLAT

Watchlist (imminent-buy clusters, p > 80%, not yet long)

ticker	cluster	theme	prob_buy	cluster_nss	basket_weight	basket_cor	sticky	etf_signal
MCHI	cluster_13	China Equity (13/15)	1.00	0.858	1.376	0.98	no	FLAT
GUNR	cluster_12	Materials (13/17)	1.00	0.550	0.681	0.97	no	FLAT
EWJ	cluster_20	Japan Equity (7/9)	1.00	0.947	0.481	0.97	no	LONG
VTEB	cluster_16	Municipal Bond (13/13)	1.00	0.244	0.468	0.97	no	FLAT
IEFA	cluster_1	Mixed: VEA, SPDW, SCHF	1.00	0.622	0.421	0.99	no	FLAT
ICSH	cluster_23	Cash / Ultra-Short Fixed In...	1.00	1.000	0.418	0.84	no	FLAT
QQQ	cluster_3	Technology (34/63)	1.00	1.000	0.398	0.98	no	LONG
VGT	cluster_3	Technology (34/63)	1.00	1.000	0.396	0.98	no	FLAT
MUB	cluster_16	Municipal Bond (13/13)	1.00	0.244	0.380	0.97	no	LONG
SHV	cluster_23	Cash / Ultra-Short Fixed In...	1.00	1.000	0.377	0.84	no	FLAT
BBJP	cluster_20	Japan Equity (7/9)	1.00	0.947	0.365	0.97	no	LONG
VEA	cluster_1	Mixed: VEA, SPDW, SCHF	1.00	0.622	0.339	0.99	no	LONG
SUB	cluster_16	Municipal Bond (13/13)	1.00	0.244	0.332	0.97	no	LONG
IGOV	cluster_24	Currency (4/6)	1.00	0.873	0.315	0.98	no	LONG
FXF	cluster_24	Currency (4/6)	1.00	0.873	0.306	0.98	no	LONG
BWX	cluster_24	Currency (4/6)	1.00	0.873	0.269	0.98	no	LONG
VUG	cluster_3	Technology (34/63)	1.00	1.000	0.198	0.98	no	FLAT
MINT	cluster_23	Cash / Ultra-Short Fixed In...	1.00	1.000	0.176	0.84	no	FLAT
VXUS	cluster_1	Mixed: VEA, SPDW, SCHF	1.00	0.622	0.172	0.99	no	LONG
COPX	cluster_12	Materials (13/17)	1.00	0.550	0.161	0.97	no	LONG
GBIL	cluster_23	Cash / Ultra-Short Fixed In...	1.00	1.000	0.118	0.84	no	FLAT
VPL	cluster_20	Japan Equity (7/9)	1.00	0.947	0.072	0.97	no	LONG
URA	cluster_12	Materials (13/17)	1.00	0.550	0.069	0.97	no	LONG
KWEB	cluster_13	China Equity (13/15)	1.00	0.858	-0.038	0.98	no	FLAT
BIL	cluster_23	Cash / Ultra-Short Fixed In...	1.00	1.000	-0.040	0.84	no	FLAT
FXI	cluster_13	China Equity (13/15)	1.00	0.858	-0.395	0.98	no	FLAT
ILF	cluster_18	Mixed: ILF, EWZ, ECH	0.98	0.072	0.382	0.93	no	FLAT
EPI	cluster_25	India Equity (3/4)	0.98	0.084	0.256	1.00	no	FLAT

INDA	cluster_25	India Equity (3/4)	0.98	0.084	0.250	1.00	no	FLAT
PIN	cluster_25	India Equity (3/4)	0.98	0.084	0.249	1.00	no	FLAT
INDY	cluster_25	India Equity (3/4)	0.98	0.084	0.246	1.00	no	FLAT
ECH	cluster_18	Mixed: ILF, EWZ, ECH	0.98	0.072	0.158	0.93	no	LONG
DBA	cluster_18	Mixed: ILF, EWZ, ECH	0.98	0.072	0.056	0.93	no	FLAT
EWZ	cluster_18	Mixed: ILF, EWZ, ECH	0.98	0.072	0.047	0.93	no	FLAT

Cluster Overview

change shows the signal-state shift vs the prior run. dNSS is today's NSS minus the prior run's.

	id	name	category	basket	n	signal	change	days	cum_return	nss	dNSS
3	cluster_3	Technology (34/63)	Technology	QQQ+VUG+VGT (cor=0.98)	63	FLAT	shifted			1.000	+0.585
8	cluster_8	Emerging Markets Equity (19...	Emerging Markets ...	IEMG+VWO+EEM (cor=0.99)	25	LONG	shifted	21	9.9%	1.000	+0.595
22	cluster_22	Mixed: SRLN, BKLN, FTSL	Mixed	FLOT+BKLN+SRLN+FLRN+FLTR (c...	9	LONG	shifted	24	1.2%	1.000	+0.186
23	cluster_23	Cash / Ultra-Short Fixed In...	Cash / Ultra-Shor...	BIL+SHV+MINT+GBIL+ICSH (cor...	9	FLAT	shifted			1.000	+0.000
4	cluster_4	Mixed: IWB, ITOT, SCHB	Mixed	VOO+IVV+SPY (cor=0.99)	61	LONG	shifted	20	5.9%	0.987	+1.183
20	cluster_20	Japan Equity (7/9)	Japan Equity	EWJ+BBJP+VPL (cor=0.97)	9	FLAT	shifted			0.947	-0.014
6	cluster_6	Mixed: IWD, SCHV, VONV	Mixed	VTV+VIG+VO (cor=1.00)	54	LONG	shifted	20	2.6%	0.919	-0.070
21	cluster_21	Broad Commodities (6/9)	Broad Commodities	PDBC+FTGC+DBC (cor=0.98)	9	LONG	shifted	84	47.8%	0.918	+0.102
24	cluster_24	Currency (4/6)	Currency	BWX+IGOV+FXF (cor=0.98)	6	FLAT	shifted			0.873	-0.127
13	cluster_13	China Equity (13/15)	China Equity	KWEB+MCHI+FXI (cor=0.98)	15	FLAT	shifted			0.858	+1.200
2	cluster_2	Mixed: VB, SCHM, IJH	Mixed	IJH+IWM+VB (cor=0.99)	66	LONG	shifted	20	3.7%	0.849	+0.110
14	cluster_14	Real Estate (14/14)	Real Estate	VNQ+SCHH+XLRE (cor=1.00)	14	LONG	shifted	21	4.2%	0.816	+0.852
10	cluster_10	U.S. Dividend Equity (11/20)	U.S. Dividend Equ...	SCHD+DVY+HDV (cor=0.98)	20	LONG	shifted	14	0.6%	0.787	+0.644
19	cluster_19	Inflation-Protected Fixed I...	Inflation-Protect...	VTIP+SCHP+STIP (cor=0.99)	10	LONG	shifted	22	0.8%	0.787	+0.249
1	cluster_1	Mixed: VEA, SPDW, SCHF	Mixed	VEA+IEFA+VXUS (cor=0.99)	82	FLAT	shifted			0.622	-0.378
12	cluster_12	Materials (13/17)	Materials	URA+GUNR+COPX (cor=0.97)	17	FLAT	shifted			0.550	+0.517
9	cluster_9	High Yield Bond (10) + Pref...	High Yield Bond	HYG+EMB+PFF (cor=0.99)	22	LONG	shifted	20	0.7%	0.489	+1.459
7	cluster_7	Mixed: VBR, MDYV, IVOV	Mixed	IJR+XLF+VBR (cor=1.00)	48	LONG	shifted	21	2.9%	0.380	-0.619
16	cluster_16	Municipal Bond (13/13)	Municipal Bond	MUB+VTEB+SUB (cor=0.97)	13	FLAT	shifted			0.244	-0.051
25	cluster_25	India Equity (3/4)	India Equity	INDA+INDY+EPI+PIN (cor=1.00)	4	FLAT	shifted			0.084	-0.916
18	cluster_18	Mixed: ILF, EWZ, ECH	Mixed	EWZ+ILF+DBA+ECH (cor=0.93)	11	FLAT	shifted			0.072	-0.138
11	cluster_11	Healthcare (16/17)	Healthcare	XLV+VHT+XBI (cor=0.99)	17	LONG	shifted	3	-2.0%	0.021	-0.188
5	cluster_5	Mixed: BND, AGG, SCHZ	Mixed	BND+AGG+BNDX (cor=1.00)	58	LONG	shifted	0	0.0%	-0.177	-1.089
15	cluster_15	Mixed: DBP, GLTR, GDXJ	Mixed	GLD+IAU+SLV (cor=0.95)	13	FLAT	shifted			-0.193	-0.964
17	cluster_17	Consumer Staples (6/11)	Consumer Staples	XLU+XLP+VPU (cor=0.99)	11	LONG	shifted	6	-2.3%	-0.232	-1.025
26	cluster_26	Currency (1/2)	Currency	SQQQ+UUP (cor=1.00)	2	FLAT	shifted			-1.000	-1.791

Net Signal Strength (NSS)

NSS = P(buy crossover within 5 days) - P(sell crossover within 5 days). Range [-1, 1]; positive = bullish lean, negative = bearish lean.

ETF rows only (Cluster Overview already covers clusters), cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive 126d trailing return — i.e. would not be blocked by the strategy's entry filter.

Top 20 most bullish (NSS desc, abs-mom approved, cluster-deduped)

id	name	category	cluster	cluster_name	cluster_signal	prob_buy	prob_sell	nss
NXTG	First Trust Indxx NextG ETF	Technology	cluster_3	Technology (34/63)	FLAT	1	0.00	1.000
VLUE	iShares MSCI USA Value Factor E...	U.S. Value Factor E...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	1	0.00	1.000
IBDR	iShares iBonds Dec 2026 Term Co...	Target Maturity Bond	cluster_5	Mixed: BND, AGG, SCHZ	LONG	1	0.00	1.000
MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short ...	cluster_23	Cash / Ultra-Short Fixe...	FLAT	1	0.00	1.000
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed...	cluster_22	Mixed: SRLN, BKLN, FTSL	LONG	1	0.00	1.000
IHF	iShares U.S. Health Care Provid...	Healthcare	cluster_11	Healthcare (16/17)	LONG	1	0.00	1.000
EWT	iShares MSCI Taiwan ETF	Taiwan Equity	cluster_8	Emerging Markets Equity...	LONG	1	0.00	1.000
EWM	iShares MSCI Malaysia Index Fund	Malaysia Equity	cluster_18	Mixed: ILF, EWZ, ECH	FLAT	1	0.00	1.000
FV	First Trust Dorsey Wright Focus...	Momentum Equity	cluster_2	Mixed: VB, SCHM, IJH	LONG	1	0.00	1.000
EIS	iShares MSCI Israel ETF	Israel Equity	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	1	0.00	1.000
VPL	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	cluster_20	Japan Equity (7/9)	FLAT	1	0.00	1.000
SLX	VanEck Steel ETF	Materials	cluster_12	Materials (13/17)	FLAT	1	0.00	1.000
FXV	Invesco CurrencyShares Japanese...	Currency	cluster_24	Currency (4/6)	FLAT	1	0.00	1.000
FTGC	First Trust Global Tactical Com...	Broad Commodities	cluster_21	Broad Commodities (6/9)	LONG	1	0.00	1.000
XLP	State Street Consumer Staples S...	Consumer Staples	cluster_17	Consumer Staples (6/11)	LONG	1	0.00	0.999
NURE	Nuveen Short-Term REIT ETF	Real Estate	cluster_14	Real Estate (14/14)	LONG	1	0.00	0.999
SCHD	Schwab US Dividend Equity ETF	U.S. Dividend Equity	cluster_10	U.S. Dividend Equity (1...	LONG	1	0.00	0.999
PFXF	VanEck Preferred Securities ex ...	Preferred Stock	cluster_9	High Yield Bond (10) + ...	LONG	1	0.00	0.998
ENZL	iShares MSCI New Zealand ETF	New Zealand Equity	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	1	0.01	0.993
IWN	iShares Russell 2000 Value ETF	U.S. Small Cap Value	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	1	0.01	0.987

Top 20 most bearish (NSS asc, cluster-deduped)

id	name	category	cluster	cluster_name	cluster_signal	prob_buy	prob_sell	nss
IHI	iShares U.S. Medical Devices ETF	Healthcare	cluster_11	Healthcare (16/17)	LONG	0.00	1.00	-1.000
SQQQ	ProShares UltraPro Short QQQ	Leveraged / Inverse	cluster_26	Currency (1/2)	FLAT	0.00	1.00	-1.000
SPLV	Invesco S&P 500 Low Volatility ...	U.S. Low Volatility...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.00	1.00	-1.000
EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_18	Mixed: ILF, EWZ, ECH	FLAT	0.00	1.00	-0.997
IDU	iShares U.S. Utilities ETF	Utilities	cluster_17	Consumer Staples (6/11)	LONG	0.00	0.99	-0.992

LVHD	Franklin U.S. Low Volatility Hi...	U.S. Low Volatility...	cluster_10	U.S. Dividend Equity (1...	LONG	0.04	0.99	-0.952
PPA	Invesco Aerospace & Defense ETF	Industrials	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.00	0.93	-0.926
IEF	iShares 7-10 Year Treasury Bond...	Fixed Income	cluster_5	Mixed: BND, AGG, SCHZ	LONG	0.04	0.96	-0.926
MOO	VanEck Agribusiness ETF	Consumer Staples	cluster_12	Materials (13/17)	FLAT	0.05	0.86	-0.816
GOAU	US Global GO Gold and Precious ...	Materials	cluster_15	Mixed: DBP, GLTR, GDXJ	FLAT	0.10	0.88	-0.787
EFAV	iShares MSCI EAFE Min Vol Facto...	International Equity	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.22	0.94	-0.724
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_16	Municipal Bond (13/13)	FLAT	0.26	0.96	-0.697
ESPO	VanEck Video Gaming and eSports...	Consumer Discretion...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.40	0.97	-0.564
KIE	State Street SPDR S&P Insurance...	Financials	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.56	1.00	-0.436
CHIQ	Global X MSCI China Consumer Di...	China Equity	cluster_13	China Equity (13/15)	FLAT	0.57	0.90	-0.333
INDY	iShares India 50 ETF	India Equity	cluster_25	India Equity (3/4)	FLAT	0.67	0.98	-0.306
IBUY	Amplify Online Retail ETF	Consumer Discretion...	cluster_3	Technology (34/63)	FLAT	0.87	0.98	-0.110
ANGL	VanEck Fallen Angel High Yield ...	High Yield Bond	cluster_9	High Yield Bond (10) + ...	LONG	1.00	0.95	0.053
LTPZ	Pimco 15 Year U.S. TIPS Index E...	Inflation-Protected...	cluster_19	Inflation-Protected Fix...	LONG	0.87	0.58	0.292
FXE	Invesco CurrencyShares Euro Cur...	Currency	cluster_24	Currency (4/6)	FLAT	1.00	0.65	0.346

Cluster Details**Cluster 1 – Mixed: VEA, SPDW, SCHF****Members:** 82 **Var explained:** 0.839 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
International Equity	21	0.2686
Developed ex-US Equity	9	0.1179
International Dividend Equity	6	0.0748
Global ex-US Equity	5	0.0656
Europe Equity	4	0.0518
Alternatives	4	0.0464
Eurozone Equity	3	0.0372
Asia-Pacific Equity	2	0.0242
Real Estate	2	0.0225
Global Equity	2	0.0222
Emerging Markets Fixed Income	2	0.0193
Currency	2	0.0181
Commodity	1	0.0132
Unknown	1	0.0130
Global ex-US Small Cap Equity	1	0.0129
Germany Equity	1	0.0124
France Equity	1	0.0124
United Kingdom Equity	1	0.0123
Italy Equity	1	0.0122
Europe Financials	1	0.0120
Sweden Equity	1	0.0120
Netherlands Equity	1	0.0120
Australia Equity	1	0.0119
Spain Equity	1	0.0116
Canada Equity	1	0.0114
Switzerland Equity	1	0.0114
Belgium Equity	1	0.0113
Austria Equity	1	0.0112
Poland Equity	1	0.0104
South Africa Equity	1	0.0099
New Zealand Equity	1	0.0091
Mexico Equity	1	0.0089

Replication basket – VEA+IEFA+VXUS (cor=0.99) basket lead at lag 1: -0.002

ticker	weight_ols	name	category	etf_signal
VEA	0.339	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG
IEFA	0.421	iShares Core MSCI EAFE ETF	Developed ex-US Equity	FLAT
VXUS	0.172	Vanguard Total International Stock ETF	Global ex-US Equity	LONG

Members

	ticker	weight	name	category	etf_signal
2	EFA	0.0133	iShares MSCI EAFE ETF	Developed ex-US Equity	FLAT
3	VEA	0.0133	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG
63	IEFA	0.0133	iShares Core MSCI EAFE ETF	Developed ex-US Equity	FLAT
64	SCHF	0.0133	Schwab International Equity ETF	Developed ex-US Equity	LONG
65	SPDW	0.0133	State Street SPDR Portfolio Developed W...	Developed ex-US Equity	LONG
21	DWM	0.0132	WisdomTree International Equity Fund	International Dividend E...	FLAT
34	TLTD	0.0132	FlexShares Morningstar Developed Market...	International Equity	FLAT
49	GSIE	0.0132	Goldman Sachs ActiveBeta International ...	Commodity	FLAT
52	ESGD	0.0132	iShares ESG Aware MSCI EAFE ETF	International Equity	FLAT
28	PXF	0.0131	Invesco RAFI Developed Markets ex-U.S. ...	International Equity	LONG
38	ACWX	0.0131	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	LONG
39	CWI	0.0131	State Street SPDR MSCI ACWI ex-US ETF	Global ex-US Equity	LONG
41	INTF	0.0131	iShares International Equity Factor ETF	International Equity	FLAT
42	IQLT	0.0131	iShares MSCI Intl Quality Factor ETF	International Equity	FLAT
44	QEFA	0.0131	State Street SPDR MSCI EAFE StrategicFa...	International Equity	FLAT
58	IEUR	0.0131	iShares Core MSCI Europe ETF	Europe Equity	FLAT
59	IXUS	0.0131	iShares Core MSCI Total International S...	Global ex-US Equity	LONG
60	VEU	0.0131	Vanguard FTSE All World Ex US ETF	Global ex-US Equity	LONG
61	VGK	0.0131	Vanguard FTSEEuropean ETF	Europe Equity	FLAT
62	VXUS	0.0131	Vanguard Total International Stock ETF	Global ex-US Equity	LONG
6	IEV	0.0130	iShares Europe ETF	Europe Equity	FLAT
32	EFV	0.0130	iShares MSCI EAFE Value ETF	Developed ex-US Equity	FLAT
35	JPIN	0.0130	JPMorgan Diversified Return Internation...	International Equity	FLAT
46	RODM	0.0130	Hartford Multifactor Developed Markets ...	International Equity	FLAT
47	VYMI	0.0130	Vanguard International High Dividend Yi...	International Dividend E...	LONG
48	FNDF	0.0130	Schwab Fundamental International Equity...	International Equity	LONG
56	SCHC	0.0130	Schwab International Small-Cap Equity E...	Developed ex-US Equity	LONG
75	FEU	0.0130			FLAT
24	IQDF	0.0129	FlexShares International Quality Divide...	International Equity	LONG
31	SCZ	0.0129	iShares MSCI EAFE Small-Cap ETF	Developed ex-US Equity	LONG
33	FNDC	0.0129	Schwab Fundamental International Small ...	International Equity	LONG
57	VSS	0.0129	Vanguard FTSE All-Wld ex-US SmCp Idx ETF	Global ex-US Small Cap E...	LONG
8	DLS	0.0128	WisdomTree International SmallCap Fund	International Equity	LONG
19	EZU	0.0128	iShares MSCI Eurozone ETF	Eurozone Equity	FLAT

43	IVLU	0.0128	iShares MSCI Intl Value Factor ETF	International Equity	FLAT
72	HFXI	0.0128	NYLI FTSE International Equity Currency...	International Equity	LONG
29	EFG	0.0127	iShares MSCI EAFE Growth ETF	Developed ex-US Equity	FLAT
45	FEZ	0.0127	State Street SPDR EURO STOXX 50 ETF	Eurozone Equity	FLAT
54	VIGI	0.0127	Vanguard International Dividend Appreci...	International Dividend E...	LONG
4	FEP	0.0126	First Trust Europe AlphaDEX Fund	Europe Equity	LONG
30	GWX	0.0126	State Street SPDR S&P International Sma...	International Equity	LONG
40	IMTM	0.0126	iShares MSCI Intl Momentum Factor ETF	International Equity	LONG
1	EWG	0.0124	iShares MSCI Germany Index Fund	Germany Equity	LONG
5	VIDI	0.0124	Vident International Equity Strategy ETF	International Equity	LONG
16	EWQ	0.0124	iShares MSCI France Index Fund	France Equity	FLAT
20	IDV	0.0124	iShares International Select Dividend E...	International Dividend E...	LONG
53	HDEF	0.0124	Xtrackers MSCI EAFE High Dividend Yield...	International Equity	LONG
17	EWU	0.0123	iShares MSCI United Kingdom ETF	United Kingdom Equity	LONG
50	IDLV	0.0123	Invesco S&P International Developed Low...	International Equity	LONG
79	IPKW	0.0123	Invesco International BuyBack Achievers...	International Equity	LONG
82	EWI	0.0122	iShares MSCI Italy ETF	Italy Equity	LONG
22	EPP	0.0121	iShares MSCI Pacific Ex-Japan Index Fund	Asia-Pacific Equity	FLAT
51	EFAV	0.0121	iShares MSCI EAFE Min Vol Factor ETF	International Equity	FLAT
23	EUFN	0.0120	iShares MSCI Europe Financials ETF	Europe Financials	FLAT
71	EWD	0.0120	iShares MSCI Sweden ETF	Sweden Equity	LONG
77	EWN	0.0120	iShares MSCI Netherlands Index Fund	Netherlands Equity	LONG
81	BBAX	0.0120	JPMorgan BetaBuilders Developed Asia Pa...	Asia-Pacific Equity	LONG
13	EWA	0.0119	iShares MSCI Australia Index Fund	Australia Equity	FLAT
27	DWX	0.0119	State Street SPDR S&P International Div...	International Dividend E...	LONG
12	HEZU	0.0118	iShares Currency Hedged MSCI Eurozone E...	Eurozone Equity	FLAT
36	DBEF	0.0118	Xtrackers MSCI EAFE Hedged Equity ETF	Alternatives	FLAT
37	HEFA	0.0118	iShares Currency Hedged MSCI EAFE ETF	Alternatives	FLAT
11	PID	0.0117	Invesco International Dividend Achiever...	International Dividend E...	LONG
15	EWI	0.0116	iShares MSCI Spain ETF	Spain Equity	FLAT
9	HEDJ	0.0114	WisdomTree Europe Hedged Equity Fund	Alternatives	FLAT
10	IHDG	0.0114	WisdomTree International Hedged Quality...	Alternatives	LONG
14	EWL	0.0114	iShares MSCI Switzerland ETF	Switzerland Equity	LONG
18	EWI	0.0114	iShares MSCI Canada Index Fund	Canada Equity	LONG
25	NFRA	0.0114	FlexShares STOXX Global Broad Infrastru...	Global Equity	LONG
55	VNQI	0.0114	Vanguard Global ex-U.S. Real Estate ETF	Real Estate	LONG
80	EWK	0.0113	iShares MSCI Belgium ETF	Belgium Equity	FLAT
76	EWO	0.0112	iShares MSCI Austria ETF	Austria Equity	LONG
7	RWX	0.0111	State Street SPDR Dow Jones Internation...	Real Estate	LONG
26	IGF	0.0108	iShares Global Infrastructure ETF	Global Equity	LONG
70	EPOL	0.0104	iShares MSCI Poland ETF	Poland Equity	FLAT
74	EZA	0.0099	iShares MSCI South Africa Index Fund	South Africa Equity	LONG
69	FXA	0.0097	Invesco CurrencyShares Australian Dolla...	Currency	LONG
66	EBND	0.0096	State Street SPDR Bloomberg Emerging Ma...	Emerging Markets Fixed I...	LONG
67	EMLC	0.0096	VanEck J. P. Morgan EM Local Currency B...	Emerging Markets Fixed I...	LONG

73	ENZL	0.0091	iShares MSCI New Zealand ETF	New Zealand Equity	LONG
68	EWX	0.0089	iShares MSCI Mexico ETF	Mexico Equity	LONG
78	FXC	0.0084	Invesco CurrencyShares Canadian Dollar ...	Currency	LONG

Cluster 2 – Mixed: VB, SCHM, IJH

Members: 66 **Var explained:** 0.848 **Signal:** LONG since 2026-04-13 (20 days, 3.7%)

Category breakdown

category	n_members	abs_weight
Industrials	15	0.2179
U.S. Small Cap Growth	6	0.0954
U.S. Mid Cap Growth	6	0.0949
Consumer Discretionary	6	0.0845
U.S. Mid Cap Equity	5	0.0813
U.S. Small Cap Equity	5	0.0803
Utilities	3	0.0428
U.S. Momentum Equity	3	0.0428
Growth Equity	2	0.0307
U.S. Small Cap Momentum Equity	2	0.0304
Momentum Equity	2	0.0296
Unknown	1	0.0163
U.S. Small Cap ESG Equity	1	0.0162
U.S. Small Cap Multifactor Equity	1	0.0161
U.S. Extended Market Equity	1	0.0161
U.S. Mid Cap Quality Equity	1	0.0158
U.S. Mid Cap Momentum Equity	1	0.0154
U.S. Small Cap Industrials	1	0.0151
Technology	1	0.0150
U.S. Micro Cap Equity	1	0.0150
IPO Equity	1	0.0143
U.S. Small Cap Consumer Discretionary	1	0.0142

Replication basket – IJH+IWM+VB (cor=0.99) basket lead at lag 1: -0.006

ticker	weight_ols	name	category	etf_signal
IJH	0.331	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	LONG

IWM	0.067	iShares Russell 2000 Index Fund	U.S. Small Cap Equity	LONG
VB	0.582	Vanguard Small-Cap ETF	U.S. Small Cap Equity	LONG

Members

	ticker	weight	name	category	etf_signal
23	SCHM	0.0163	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	LONG
29	MDY	0.0163	State Street SPDR S&P MIDCAP 400 ETF Tr...	U.S. Mid Cap Equity	LONG
37	IVOO	0.0163	Vanguard S&P Mid-Cap 400 ETF		LONG
42	IJH	0.0163	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	LONG
43	SPMD	0.0163	State Street SPDR Portfolio S&P 400 Mid...	U.S. Mid Cap Equity	LONG
44	VB	0.0163	Vanguard Small-Cap ETF	U.S. Small Cap Equity	LONG
20	JHMM	0.0162	John Hancock Multifactor Mid Cap ETF	U.S. Mid Cap Equity	LONG
26	NUSC	0.0162	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG Equity	LONG
31	IJK	0.0162	iShares S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
33	IVOG	0.0162	Vanguard S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
34	MDYG	0.0162	State Street SPDR S&P 400 Mid Cap Growt...	U.S. Mid Cap Growth	LONG
45	SCHA	0.0162	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
27	SMLF	0.0161	iShares U.S. Small-Cap Equity Factor ETF	U.S. Small Cap Multifact...	LONG
41	VXF	0.0161	Vanguard Extended Market ETF	U.S. Extended Market Equ...	LONG
4	IWM	0.0160	iShares Russell 2000 Index Fund	U.S. Small Cap Equity	LONG
30	IJT	0.0160	iShares S&P SmallCap 600 Growth ETF	U.S. Small Cap Growth	LONG
32	VIOG	0.0160	Vanguard S&P Small-Cap 600 Growth ETF	U.S. Small Cap Growth	LONG
35	SLYG	0.0160	State Street SPDR S&P 600 Small Cap Gro...	U.S. Small Cap Growth	LONG
38	VTWO	0.0160	Vanguard Russell 2000 ETF	U.S. Small Cap Equity	LONG
40	ISCG	0.0159	iShares Morningstar Small-Cap Growth ETF	Growth Equity	LONG
15	FXR	0.0158	First Trust Industrials AlphaDEX	Industrials	LONG
28	IWO	0.0158	iShares Russell 2000 Growth Fund	U.S. Small Cap Growth	LONG
36	VTWG	0.0158	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Growth	LONG
39	VBK	0.0158	Vanguard Small-Cap Growth ETF	U.S. Small Cap Growth	LONG
54	PSC	0.0158	Principal U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
61	XMHQ	0.0158	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Quality Equ...	LONG
14	IMCG	0.0157	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
21	VIS	0.0157	Vanguard Industrials ETF	Industrials	LONG
19	IYJ	0.0156	iShares U.S. Industrials ETF	Industrials	LONG
22	FIDU	0.0156	Fidelity MSCI Industrials Index ETF	Industrials	LONG
60	PAVE	0.0155	Global X U.S. Infrastructure Developmen...	Industrials	LONG
10	XMMO	0.0154	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Momentum Eq...	LONG
11	IWP	0.0153	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	FLAT
13	VOT	0.0153	Vanguard Mid-Cap Growth ETF	U.S. Mid Cap Growth	FLAT
46	FXD	0.0153	First Trust Cons. Discret. AlphaDEX	Consumer Discretionary	LONG
62	VFMO	0.0153	Vanguard U.S. Momentum Factor ETF	U.S. Momentum Equity	LONG
66	XSMO	0.0153	Invesco S&P SmallCap Momentum ETF	U.S. Small Cap Momentum ...	LONG

1	XLI	0.0152	State Street Industrial Select Sector S...	Industrials	LONG
59	PSCI	0.0151	Invesco S&P SmallCap Industrials ETF	U.S. Small Cap Industria...	LONG
6	FTC	0.0150	First Trust Large Cap Growth AlphaDEX F...	Technology	LONG
24	DWAS	0.0150	Invesco Dorsey Wright SmallCap Momentum...	U.S. Small Cap Momentum ...	LONG
25	IWC	0.0150	iShares Microcap ETF	U.S. Micro Cap Equity	LONG
53	RSPD	0.0150	Invesco S&P 500 Equal Weight Consumer D...	Consumer Discretionary	FLAT
7	PDP	0.0149	Invesco Dorsey Wright Momentum ETF	Momentum Equity	LONG
9	RPG	0.0148	Invesco S&P 500 Pure Growth ETF	Growth Equity	LONG
18	FIW	0.0148	First Trust Water ETF	Utilities	FLAT
48	FTXR	0.0148	First Trust Nasdaq Transportation ETF	Industrials	LONG
5	FV	0.0147	First Trust Dorsey Wright Focus 5 ETF	Momentum Equity	LONG
63	PKB	0.0147	Invesco Building & Construction ETF	Industrials	LONG
64	PRN	0.0146	Invesco Dorsey Wright Industrials Momen...	Industrials	LONG
16	PHO	0.0145	Invesco Water Resources ETF	Utilities	FLAT
8	FPX	0.0143	First Trust US Equity Opportunities ETF	IPO Equity	LONG
57	XTN	0.0143	State Street SPDR S&P Transportation ETF	Industrials	FLAT
47	PEJ	0.0142	Invesco Leisure and Entertainment ETF	Consumer Discretionary	FLAT
65	PSCD	0.0142	Invesco S&P SmallCap Consumer Discretio...	U.S. Small Cap Consumer ...	FLAT
52	IYT	0.0141	iShares U.S. Transportation ETF	Industrials	LONG
2	XRT	0.0139	State Street SPDR S&P Retail ETF	Consumer Discretionary	FLAT
58	QMOM	0.0138	Alpha Architect U.S. Quantitative Momen...	U.S. Momentum Equity	LONG
12	MTUM	0.0137	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	LONG
56	XHB	0.0137	State Street SPDR S&P Homebuilders ETF	Consumer Discretionary	LONG
17	CGW	0.0135	Invesco S&P Global Water Index ETF	Utilities	FLAT
55	XAR	0.0135	State Street SPDR S&P Aerospace & Defen...	Industrials	FLAT
49	PPA	0.0133	Invesco Aerospace & Defense ETF	Industrials	FLAT
3	JETS	0.0127	U.S. Global Jets ETF	Industrials	LONG
50	ITA	0.0125	iShares U.S. Aerospace & Defense ETF	Industrials	FLAT
51	ITB	0.0124	iShares U.S. Home Construction ETF	Consumer Discretionary	FLAT

Cluster 3 – Technology (34/63)

Members: 63 **Var explained:** 0.840 **Signal:** FLAT

Category breakdown

category	n_members	abs_weight
Technology	34	0.5348
U.S. Large Cap Growth	11	0.1854
U.S. Equity	2	0.0333

Growth Equity	2	0.0332
Unknown	2	0.0315
Convertible Bond	2	0.0306
Financials	2	0.0298
U.S. Mega Cap Growth	1	0.0167
U.S. Equal Weight Equity	1	0.0167
U.S. Small Cap Technology	1	0.0153
IPO Equity	1	0.0152
Options Income	1	0.0152
Trend-Following Equity	1	0.0148
Consumer Discretionary	1	0.0147
U.S. Trend-Following Equity	1	0.0128

Replication basket – QQQ+VUG+VGT (cor=0.98) basket lead at lag 1: -0.010

ticker	weight_ols	name	category	etf_signal
QQQ	0.398	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	LONG
VUG	0.198	Vanguard Growth ETF	U.S. Large Cap Growth	FLAT
VGT	0.396	Vanguard Information Tech ETF	Technology	FLAT

Members

	ticker	weight	name	category	etf_signal
14	QQQ	0.0170	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	LONG
35	IGM	0.0170	iShares Expanded Tech Sector ETF	Technology	FLAT
40	ONEQ	0.0170	Fidelity Nasdaq Composite Index ETF	U.S. Equity	LONG
50	ILCG	0.0170	iShares Morningstar Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
29	XNTK	0.0169	State Street SPDR NYSE Technology ETF	Technology	FLAT
39	NULG	0.0169	Nuveen ESG Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
43	IWF	0.0169	iShares Russell 1000 Growth Fund	U.S. Large Cap Growth	LONG
45	VGT	0.0169	Vanguard Information Tech ETF	Technology	FLAT
47	FTEC	0.0169	Fidelity MSCI Information Technology In...	Technology	FLAT
48	VONG	0.0169	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Growth	LONG
52	SCHG	0.0169	Schwab U.S. Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
54	VUG	0.0169	Vanguard Growth ETF	U.S. Large Cap Growth	FLAT
8	AIQ	0.0168	Global X Artificial Intelligence & Tech...	Technology	FLAT
18	QTEC	0.0168	First Trust NASDAQ-100-Technology Secto...	Technology	FLAT
24	FXL	0.0168	First Trust Technology AlphaDEX	Technology	FLAT
37	IYW	0.0168	iShares U.S. Technology ETF	Technology	FLAT
44	IVW	0.0168	iShares S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
51	IUSG	0.0168	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Growth	LONG

17	QQEW	0.0167	First Trust Nasdaq-100 Select Equal Wei...	U.S. Equal Weight Equity	FLAT
36	IXN	0.0167	iShares Global Tech ETF	Technology	LONG
42	IWY	0.0167	iShares Russell Top 200 Growth ETF	Growth Equity	LONG
46	VOOG	0.0167	Vanguard S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
49	MGK	0.0167	Vanguard Mega Cap Growth ETF	U.S. Mega Cap Growth	FLAT
53	SPYG	0.0167	State Street SPDR Portfolio S&P 500 Gro...	U.S. Large Cap Growth	LONG
1	XLK	0.0166	State Street Technology Select Sector S...	Technology	FLAT
21	XT	0.0165	iShares Future Exponential Technologies...	Technology	LONG
34	PWB	0.0165	Invesco Large Cap Growth ETF	Growth Equity	LONG
55	SPHB	0.0162	Invesco S&P 500 High Beta ETF	U.S. Equity	LONG
20	TDIV	0.0160	First Trust NASDAQ Technology Dividend ...	Technology	LONG
28	FDN	0.0160	First Trust DJ Internet Index Fund	Technology	FLAT
60	FIVG	0.0160			FLAT
6	BOTZ	0.0159	Global X Robotics & Artificial Intellig...	Technology	LONG
7	ROBO	0.0159	ROBO Global Robotics and Automation Ind...	Technology	LONG
63	QTUM	0.0159	Defiance Quantum ETF	Technology	LONG
27	PNQI	0.0158	Invesco Nasdaq Internet ETF	Technology	FLAT
5	SKYY	0.0157	First Trust Cloud Computing ETF	Technology	FLAT
2	SOXX	0.0156	iShares PHLX SOX Semiconductor Sector I...	Technology	LONG
3	SMH	0.0156	VanEck Semiconductor ETF	Technology	LONG
4	IGV	0.0155	iShares Expanded Tech-Software Sector E...	Technology	FLAT
22	XSD	0.0155	State Street SPDR S&P Semiconductor ETF	Technology	LONG
38	CWB	0.0155	State Street SPDR Bloomberg Convertible...	Convertible Bond	LONG
61	PBS	0.0155			FLAT
62	PTF	0.0155	Invesco Dorsey Wright Technology Moment...	Technology	LONG
13	ARKQ	0.0154	ARK Autonomous Technology & Robotics ETF	Technology	LONG
15	NXTG	0.0154	First Trust Indxx NextG ETF	Technology	LONG
19	PSI	0.0153	Invesco Semiconductors ETF	Technology	LONG
57	PSCT	0.0153	Invesco S&P SmallCap Information Techno...	U.S. Small Cap Technology	LONG
26	IPO	0.0152	Renaissance IPO ETF	IPO Equity	FLAT
31	FINX	0.0152	Global X FinTech ETF	Financials	FLAT
33	QYLD	0.0152	Global X NASDAQ 100 Covered Call ETF	Options Income	LONG
59	FTXL	0.0152	First Trust Nasdaq Semiconductor ETF	Technology	LONG
12	ARKW	0.0151	ARK Next Generation Internet ETF	Technology	FLAT
41	ICVT	0.0151	iShares Convertible Bond ETF	Convertible Bond	LONG
11	ARKK	0.0148	ARK Innovation ETF	Technology	FLAT
25	CIBR	0.0148	First Trust NASDAQ Cybersecurity ETF	Technology	FLAT
32	PTNQ	0.0148	Pacer Trendpilot 100 ETF	Trend-Following Equity	LONG
23	IBUY	0.0147	Amplify Online Retail ETF	Consumer Discretionary	FLAT
10	HACK	0.0146	Amplify Cybersecurity ETF	Technology	FLAT
30	IPAY	0.0146	Amplify Digital Payments ETF	Financials	FLAT
16	PRNT	0.0144	3D Printing (The) ETF	Technology	FLAT
9	BLOK	0.0137	Amplify Blockchain Technology ETF	Technology	FLAT
56	PTLC	0.0128	Pacer Trendpilot US Large Cap ETF	U.S. Trend-Following Equ...	LONG
58	BLCN	0.0123	Siren NexGen Economy ETF	Technology	FLAT

Cluster 4 – Mixed: IWB, ITOT, SCHB**Members:** 61 **Var explained:** 0.892 **Signal:** LONG since 2026-04-13 (20 days, 5.9%)**Category breakdown**

category	n_members	abs_weight
U.S. Large Cap Equity	9	0.1558
U.S. Total Market Equity	7	0.1214
Global Equity	7	0.1191
Consumer Discretionary	6	0.0914
Asset Allocation	4	0.0631
Communication Services	4	0.0592
U.S. Mega Cap Equity	3	0.0510
U.S. Quality / Factor Equity	3	0.0505
U.S. Multifactor Equity	3	0.0504
U.S. ESG Equity	2	0.0345
U.S. Equity	2	0.0345
Unknown	2	0.0312
Commodity	1	0.0173
Large Cap Value Equities	1	0.0172
Large Cap Blend Equities	1	0.0171
Dividend Equity	1	0.0168
U.S. Dividend Equity	1	0.0165
Alternatives	1	0.0149
International IPO Equity	1	0.0133
Israel Equity	1	0.0126
Singapore Equity	1	0.0121

Replication basket – VOO+IVV+SPY (cor=0.99) basket lead at lag 1: -0.005

ticker	weight_ols	name	category	etf_signal
VOO	0.804	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG
IVV	0.309	iShares Core S&P 500 ETF	U.S. Large Cap Equity	LONG
SPY	-0.154	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity	LONG

Members

	ticker	weight	name	category	etf_signal
37	IWB	0.0174	iShares Russell 1000 ETF	U.S. Large Cap Equity	LONG

49	ITOT	0.0174	iShares Core S&P Total U.S. Stock Marke...	U.S. Total Market Equity	LONG
51	SCHB	0.0174	Schwab U.S. Broad Market ETF	U.S. Total Market Equity	LONG
4	SPY	0.0173	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity	LONG
12	URTH	0.0173	iShares MSCI World ETF	Global Equity	LONG
30	IWV	0.0173	iShares Russell 3000 Fund	U.S. Total Market Equity	LONG
31	IYY	0.0173	iShares Dow Jones U.S. ETF	U.S. Total Market Equity	LONG
34	SPYX	0.0173	State Street SPDR S&P 500 Fossil Fuel R...	U.S. Equity	LONG
36	ESGU	0.0173	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	LONG
42	VTHR	0.0173	Vanguard Russell 3000 ETF	U.S. Total Market Equity	LONG
43	GSLC	0.0173	Goldman Sachs ActiveBeta U.S. Large Cap...	Commodity	LONG
45	VONE	0.0173	Vanguard Russell 1000 ETF	U.S. Large Cap Equity	LONG
47	VV	0.0173	Vanguard Large-Cap ETF	U.S. Large Cap Equity	LONG
48	ILCB	0.0173	iShares Morningstar Large-Cap ETF	U.S. Large Cap Equity	LONG
50	IVV	0.0173	iShares Core S&P 500 ETF	U.S. Large Cap Equity	LONG
52	SCHX	0.0173	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equity	LONG
53	SPLG	0.0173			LONG
54	SPTM	0.0173	State Street SPDR Portfolio S&P 1500 Co...	U.S. Total Market Equity	LONG
55	VOO	0.0173	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG
56	VTI	0.0173	Vanguard Total Stock Market ETF	U.S. Total Market Equity	LONG
26	JHML	0.0172	John Hancock Multifactor Large Cap ETF	U.S. Large Cap Equity	LONG
29	SUSA	0.0172	iShares ESG Optimized MSCI USA ETF	U.S. ESG Equity	LONG
38	IWL	0.0172	iShares Russell Top 200 ETF	U.S. Equity	LONG
44	EPS	0.0172	WisdomTree U.S. LargeCap Fund	Large Cap Value Equities	LONG
46	MGC	0.0172	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	LONG
7	ACWI	0.0171	iShares MSCI ACWI ETF	Global Equity	LONG
13	CRBN	0.0171	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity	LONG
15	VT	0.0171	Vanguard Total World Stock Index ETF	Global Equity	LONG
22	TILT	0.0171	FlexShares Mornigstar US Market Factors...	U.S. Multifactor Equity	LONG
28	DSI	0.0171	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equities	LONG
32	LRGF	0.0171	iShares U.S. Equity Factor ETF	U.S. Multifactor Equity	LONG
8	AOA	0.0170	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation	LONG
14	SPGM	0.0170	State Street SPDR Portfolio MSCI Global...	Global Equity	LONG
33	OEF	0.0170	iShares S&P 100 Fund	U.S. Mega Cap Equity	LONG
39	QUAL	0.0170	iShares MSCI USA Quality Factor ETF	U.S. Quality / Factor Eq...	LONG
24	QDF	0.0168	FlexShares Quality Dividend Index Fund	Dividend Equity	LONG
40	QUS	0.0168	State Street SPDR MSCI USA StrategicFac...	U.S. Quality / Factor Eq...	LONG
23	IOO	0.0167	iShares Global 100 ETF	Global Equity	LONG
25	CATH	0.0167	Global X S&P 500 Catholic Values ETF	Global Equity	LONG
35	XLG	0.0167	Invesco S&P 500 Top 50 ETF	U.S. Mega Cap Equity	FLAT
41	SPHQ	0.0166	Invesco S&P 500 Quality ETF	U.S. Quality / Factor Eq...	LONG
11	AOR	0.0165	iShares Core 60/40 Balanced Allocation ...	Asset Allocation	LONG
27	DGRW	0.0165	WisdomTree U.S. Quality Dividend Growth...	U.S. Dividend Equity	LONG
17	IYC	0.0162	iShares U.S. Consumer Discretionary ETF	Consumer Discretionary	LONG
60	OMFL	0.0161	Invesco Russell 1000 Dynamic Multifacto...	U.S. Multifactor Equity	LONG
21	FDIS	0.0159	Fidelity MSCI Consumer Discretionary In...	Consumer Discretionary	LONG

18	VCR	0.0158	Vanguard Consumer Discretion ETF	Consumer Discretionary	LONG
1	XLY	0.0156	State Street Consumer Discretionary Sel...	Consumer Discretionary	LONG
10	AOM	0.0155	iShares Core 40/60 Moderate Allocation ...	Asset Allocation	LONG
19	VOX	0.0153	Vanguard Communication Services ETF	Communication Services	LONG
20	FCOM	0.0153	Fidelity MSCI Communication Services In...	Communication Services	LONG
2	XLC	0.0149	State Street Communication Services Sel...	Communication Services	LONG
5	QAI	0.0149	NYLI Hedge Multi-Strategy Tracker ETF	Alternatives	LONG
59	RTH	0.0147	VanEck Retail ETF	Consumer Discretionary	LONG
9	AOK	0.0142	iShares Core 30/70 Conservative Allocat...	Asset Allocation	LONG
61	IGN	0.0139			FLAT
57	IYZ	0.0137	iShares U.S. Telecommunications ETF	Communication Services	FLAT
16	FPXI	0.0133	First Trust International Equity Opport...	International IPO Equity	LONG
3	ESPO	0.0132	VanEck Video Gaming and eSports ETF	Consumer Discretionary	FLAT
58	EIS	0.0126	iShares MSCI Israel ETF	Israel Equity	LONG
6	EWS	0.0121	iShares MSCI Singapore ETF	Singapore Equity	LONG

Cluster 5 – Mixed: BND, AGG, SCHZ

Members: 58 **Var explained:** 0.832 **Signal:** LONG since 2026-05-11 (0 days, 0.0%)

Category breakdown

category	n_members	abs_weight
Fixed Income	19	0.3282
Investment Grade Corporate Bond	13	0.2301
Aggregate Fixed Income	6	0.1120
Mortgage-Backed Fixed Income	6	0.0982
Active Fixed Income	5	0.0890
Cash / Ultra-Short Fixed Income	3	0.0470
International Fixed Income	2	0.0313
Target Maturity Bond	2	0.0309
Agency Fixed Income	1	0.0166
Taxable Municipal Bond	1	0.0165

Replication basket – BND+AGG+BNDX (cor=1.00) basket lead at lag 1: 0.010

ticker	weight_ols	name	category	etf_signal
BND	0.629	Vanguard Total Bond Market ETF	Aggregate Fixed Income	LONG

AGG	0.354	iShares Core U.S. Aggregate Bond ETF	Aggregate Fixed Income	LONG
BNDX	0.033	Vanguard Total International Bond ETF	International Fixed I...	LONG

Members

	ticker	weight	name	category	etf_signal
21	BIV	0.0188	Vanguard Intermediate-Term Bond ETF	Fixed Income	FLAT
26	AGG	0.0188	iShares Core U.S. Aggregate Bond ETF	Aggregate Fixed Income	LONG
28	BND	0.0188	Vanguard Total Bond Market ETF	Aggregate Fixed Income	LONG
30	SCHZ	0.0188	Schwab US Aggregate Bond ETF	Aggregate Fixed Income	LONG
20	IUSB	0.0187	iShares Core Universal USD Bond ETF	Aggregate Fixed Income	LONG
31	SPAB	0.0187	State Street SPDR Portfolio Aggregate B...	Aggregate Fixed Income	LONG
2	IEF	0.0185	iShares 7-10 Year Treasury Bond ETF	Fixed Income	FLAT
17	IGIB	0.0184	iShares 5-10 Year Investment Grade Corp...	Investment Grade Corpora...	LONG
24	VCIT	0.0184	Vanguard Intermediate-Term Corporate Bo...	Investment Grade Corpora...	LONG
7	FBND	0.0183	Fidelity Total Bond ETF	Active Fixed Income	LONG
43	SPTI	0.0183	State Street SPDR Portfolio Intermediat...	Fixed Income	FLAT
47	VGIT	0.0183	Vanguard Intermediate-Term Treasury ETF	Fixed Income	FLAT
13	AGGY	0.0182	WisdomTree Yield Enhanced U.S. Aggregat...	Aggregate Fixed Income	LONG
15	SPIB	0.0182	State Street SPDR Portfolio Intermediat...	Investment Grade Corpora...	LONG
46	SCHR	0.0182	Schwab Intermediate-Term U.S. Treasury ...	Fixed Income	FLAT
6	GTO	0.0181	Invesco Total Return Bond ETF	Active Fixed Income	LONG
11	GVI	0.0181	iShares Intermediate Government/Credit ...	Fixed Income	LONG
40	IEI	0.0181	iShares 3-7 Year Treasury Bond ETF	Fixed Income	FLAT
12	QLTA	0.0180	iShares Aaa A Rated Corporate Bond ETF	Investment Grade Corpora...	LONG
27	USIG	0.0180	iShares Broad USD Investment Grade Corp...	Investment Grade Corpora...	LONG
50	MBB	0.0180	iShares MBS ETF	Mortgage-Backed Fixed In...	LONG
51	VMBS	0.0180	Vanguard Mortgage-Backed Securities ETF	Mortgage-Backed Fixed In...	LONG
3	LQD	0.0179	iShares iBoxx \$ Investment Grade Corpor...	Investment Grade Corpora...	LONG
5	BOND	0.0179	PIMCO Active Bond Exchange-Traded Fund ...	Active Fixed Income	LONG
52	SPMB	0.0179	State Street SPDR Portfolio Mortgage Ba...	Mortgage-Backed Fixed In...	LONG
19	ISTB	0.0178	iShares Core 1-5 Year USD Bond ETF	Fixed Income	FLAT
8	FLCO	0.0177	Franklin Investment Grade Corporate ETF	Active Fixed Income	LONG
10	CORP	0.0177	PIMCO Investment Grade Corporate Bond I...	Investment Grade Corpora...	LONG
22	BLV	0.0176	Vanguard Long-Term Bond ETF	Fixed Income	FLAT
41	TLH	0.0176	iShares 10-20 Year Treasury Bond ETF	Fixed Income	FLAT
45	GOVT	0.0176	iShares U.S. Treasury Bond ETF	Fixed Income	FLAT
35	IGSB	0.0175	iShares 1-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
37	VCSH	0.0175	Vanguard Short-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
23	BSV	0.0174	Vanguard Short-Term Bond ETF	Fixed Income	LONG
49	GNMA	0.0174	iShares GNMA Bond ETF	Mortgage-Backed Fixed In...	LONG
18	IGLB	0.0172	iShares 10 Year Investment Grade Corpor...	Investment Grade Corpora...	LONG
44	SPTL	0.0172	State Street SPDR Portfolio Long Term T...	Fixed Income	FLAT

16	SPLB	0.0171	State Street SPDR Portfolio Long Term C...	Investment Grade Corpora...	LONG
25	VCLT	0.0171	Vanguard Long-Term Corporate Bond ETF	Investment Grade Corpora...	LONG
36	SLQD	0.0171	iShares 0-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
48	VGLT	0.0171	Vanguard Long-Term Treasury ETF	Fixed Income	FLAT
29	TOTL	0.0170	State Street DoubleLine Total Return Ta...	Active Fixed Income	FLAT
1	TLT	0.0169	iShares 20+ Year Treasury Bond ETF	Fixed Income	FLAT
39	AGZ	0.0166	iShares Agency Bond ETF	Agency Fixed Income	LONG
9	BAB	0.0165	Invesco Taxable Municipal Bond ETF	Taxable Municipal Bond	LONG
42	EDV	0.0163	Vanguard Extended Duration Treasury ETF	Fixed Income	FLAT
34	SPSB	0.0162	State Street SPDR Portfolio Short Term ...	Fixed Income	LONG
53	SHY	0.0162	iShares 1-3 Year Treasury Bond ETF	Fixed Income	FLAT
56	VGSH	0.0159	Vanguard Short-Term Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
57	BNDX	0.0158	Vanguard Total International Bond ETF	International Fixed Inco...	LONG
33	BSCQ	0.0157	Invesco BulletShares 2026 Corporate Bon...	Target Maturity Bond	FLAT
55	SCHO	0.0156	Schwab Short-Term U.S. Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
58	IAGG	0.0156	iShares International Aggregate Bond Fu...	International Fixed Inco...	LONG
54	SPTS	0.0155	State Street SPDR Portfolio Short Term ...	Cash / Ultra-Short Fixed...	FLAT
14	IBDR	0.0152	iShares iBonds Dec 2026 Term Corporate ...	Target Maturity Bond	FLAT
38	CMBS	0.0136	iShares CMBS Bond ETF	Mortgage-Backed Fixed In...	LONG
4	LMBS	0.0133	First Trust Low Duration Opportunities ...	Mortgage-Backed Fixed In...	FLAT
32	LDUR	0.0120	PIMCO Enhanced Low Duration Active Exch...	Fixed Income	LONG

Cluster 6 – Mixed: IWD, SCHV, VONV

Members: 54 **Var explained:** 0.922 **Signal:** LONG since 2026-04-13 (20 days, 2.6%)

Category breakdown

category	n_members	abs_weight
U.S. Dividend Equity	11	0.2040
U.S. Large Cap Value	10	0.1903
U.S. Multifactor Equity	3	0.0559
U.S. Mid Cap Equity	3	0.0556
U.S. Value Factor Equity	3	0.0553
U.S. Low Volatility Equity	3	0.0514
U.S. Fundamental Equity	2	0.0380
U.S. Equal Weight Equity	2	0.0377
U.S. Mid Cap Value	2	0.0376
U.S. Equity	2	0.0375
Value Equity	2	0.0369

U.S. Revenue Weighted Equity	1	0.0188
Large Cap Blend Equities	1	0.0188
U.S. Mega Cap Value	1	0.0188
U.S. Size Factor Equity	1	0.0187
U.S. Large Cap Equity	1	0.0184
U.S. Buyback Equity	1	0.0183
Dividend Equity	1	0.0180
U.S. Quality / Factor Equity	1	0.0179
U.S. Moat Equity	1	0.0177
U.S. ESG Equity	1	0.0176
Global Equity	1	0.0168

Replication basket – VTV+VIG+VO (cor=1.00) basket lead at lag 1: 0.012

ticker	weight_ols	name	category	etf_signal
VTV	0.546	Vanguard Value ETF	U.S. Large Cap Value	LONG
VIG	0.158	Vanguard Div Appreciation ETF	U.S. Dividend Equity	LONG
VO	0.278	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	LONG

Members

	ticker	weight	name	category	etf_signal
22	IWD	0.0192	iShares Russell 1000 Value ETF	U.S. Large Cap Value	LONG
2	RSP	0.0191	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity	LONG
9	PRF	0.0191	Invesco RAFI US 1000 ETF	U.S. Fundamental Equity	LONG
29	VONV	0.0191	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	LONG
32	ILCV	0.0191	iShares Morningstar Large-Cap Value ETF	U.S. Large Cap Value	LONG
35	SCHV	0.0191	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	LONG
12	CFA	0.0190	VictoryShares US 500 Volatility Wtd ETF	U.S. Multifactor Equity	LONG
18	FNDX	0.0190	Schwab Fundamental U.S. Large Company E...	U.S. Fundamental Equity	LONG
24	IVE	0.0190	iShares S&P 500 Value ETF	U.S. Large Cap Value	LONG
25	JPUS	0.0190	JPMorgan Diversified Return U.S. Equity...	U.S. Multifactor Equity	LONG
34	IUSV	0.0190	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	LONG
38	VTV	0.0190	Vanguard Value ETF	U.S. Large Cap Value	LONG
45	DTD	0.0190	WisdomTree U.S. Total Dividend Fund	U.S. Dividend Equity	LONG
50	DGRO	0.0190	iShares Core Dividend Growth ETF	U.S. Dividend Equity	LONG
15	NULV	0.0189	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	LONG
20	IWX	0.0189	iShares Russell Top 200 Value ETF	Value Equity	LONG
28	VOOV	0.0189	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	LONG
36	SPYV	0.0189	State Street SPDR Portfolio S&P 500 Val...	U.S. Large Cap Value	LONG
44	DLN	0.0189	WisdomTree U.S. LargeCap Dividend Fund	U.S. Dividend Equity	LONG

5	FEX	0.0188	First Trust Large Cap Core AlphaDEX Fund	Large Cap Blend Equities	LONG
10	RWL	0.0188	Invesco S&P 500 Revenue ETF	U.S. Revenue Weighted Eq...	LONG
30	MGV	0.0188	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	LONG
31	VYM	0.0188	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	LONG
39	IWS	0.0188	iShares Russell Mid-Cap Value ETF	U.S. Mid Cap Value	LONG
40	VOE	0.0188	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	LONG
42	PFM	0.0188	Invesco Dividend Achievers ETF	U.S. Dividend Equity	LONG
11	AIVL	0.0187	WisdomTree U.S. AI Enhanced Value Fund	U.S. Value Factor Equity	LONG
21	ONEV	0.0187	State Street SPDR Russell 1000 Low Vola...	U.S. Equity	LONG
26	SIZE	0.0187	iShares MSCI USA Size Factor ETF	U.S. Size Factor Equity	LONG
51	VIG	0.0187	Vanguard Div Appreciation ETF	U.S. Dividend Equity	LONG
53	JPME	0.0187	JPMorgan Diversified Return U.S. Mid Ca...	U.S. Equity	LONG
19	EQAL	0.0186	Invesco Russell 1000 Equal Weight ETF	U.S. Equal Weight Equity	LONG
33	IMCB	0.0186	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	LONG
54	JVAL	0.0186	JPMorgan U.S. Value Factor ETF	U.S. Value Factor Equity	LONG
17	FDVV	0.0185	Fidelity High Dividend ETF	U.S. Dividend Equity	LONG
23	IWR	0.0185	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	LONG
37	VO	0.0185	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	LONG
47	DIA	0.0184	State Street SPDR Dow Jones Industrial ...	U.S. Large Cap Equity	LONG
4	PKW	0.0183	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	FLAT
14	NOBL	0.0182	ProShares S&P 500 Dividend Aristocrats ...	U.S. Dividend Equity	FLAT
6	PWV	0.0181	Invesco Large Cap Value ETF	Value Equity	LONG
7	RDVY	0.0181	First Trust Rising Dividend Achievers E...	U.S. Dividend Equity	LONG
3	FVD	0.0180	First Trust VL Dividend	U.S. Dividend Equity	FLAT
16	SDY	0.0180	State Street SPDR S&P Dividend ETF	U.S. Dividend Equity	LONG
27	VLUE	0.0180	iShares MSCI USA Value Factor ETF	U.S. Value Factor Equity	LONG
43	OUSX	0.0180	ALPS O'Shares U.S. Quality Dividend ETF	Dividend Equity	LONG
41	FTCS	0.0179	First Trust Capital Strength ETF	U.S. Quality / Factor Eq...	FLAT
49	LGLV	0.0179	State Street SPDR US Large Cap Low Vola...	U.S. Low Volatility Equi...	FLAT
13	CFO	0.0178	VictoryShares US 500 Enhanced Volatilit...	U.S. Multifactor Equity	LONG
8	MOAT	0.0177	VanEck Morningstar Wide Moat ETF	U.S. Moat Equity	LONG
48	USMV	0.0176	iShares MSCI USA Min Vol Factor ETF	U.S. Low Volatility Equi...	LONG
52	ETHO	0.0176	Amplify Etho Climate Leadership U.S. ETF	U.S. ESG Equity	LONG
1	ACWV	0.0168	iShares MSCI Global Min Vol Factor ETF	Global Equity	LONG
46	SPLV	0.0159	Invesco S&P 500 Low Volatility ETF	U.S. Low Volatility Equi...	FLAT

Cluster 7 – Mixed: VBR, MDYV, IVOV**Members:** 48 **Var explained:** 0.880 **Signal:** LONG since 2026-04-10 (21 days, 2.9%)**Category breakdown**

category	n_members	abs_weight
Financials	13	0.2576
U.S. Small Cap Value	6	0.1304
U.S. Small Cap Equity	5	0.1082
U.S. Mid Cap Value	3	0.0657
Dividend Equity	2	0.0424
Leveraged / Inverse	2	0.0422
U.S. Small Cap Low Volatility	2	0.0407
U.S. Small Cap Fundamental Equity	1	0.0218
Mid Cap Value Equities	1	0.0218
U.S. Mid Cap Dividend Equity	1	0.0217
Small Cap Value Equities	1	0.0216
U.S. Small Cap Dividend Equity	1	0.0216
U.S. Mid/Small Cap Equity	1	0.0216
U.S. Small/Mid Cap Fundamental Equity	1	0.0215
Value Equity	1	0.0215
U.S. Mid Cap Equity	1	0.0214
U.S. Small/Mid Cap Equity	1	0.0213
U.S. Equity	1	0.0210
U.S. Dividend Equity	1	0.0206
U.S. Mid Cap Low Volatility	1	0.0198
Global Equity	1	0.0181
Real Estate	1	0.0173

Replication basket – IJR+XLF+VBR (cor=1.00) basket lead at lag 1: -0.003

ticker	weight_ols	name	category	etf_signal
IJR	0.363	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	LONG
XLF	0.219	State Street Financial Select Sector ...	Financials	FLAT
VBR	0.430	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	LONG

Members

	ticker	weight	name	category	etf_signal
37	VBR	0.0220	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	LONG
28	IJJ	0.0219	iShares S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	LONG
30	IVOV	0.0219	Vanguard S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	LONG
31	MDYV	0.0219	State Street SPDR S&P 400 Mid Cap Value...	U.S. Mid Cap Value	LONG
21	EZM	0.0218	WisdomTree U.S. MidCap Fund	Mid Cap Value Equities	LONG
23	FNDA	0.0218	Schwab Fundamental U.S. Small Company E...	U.S. Small Cap Fundament...	LONG

8	FYX	0.0217	First Trust Small Cap Core AlphaDEX Fund	U.S. Small Cap Equity	LONG
10	VI00	0.0217	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equity	LONG
11	IJR	0.0217	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	LONG
12	SPSM	0.0217	State Street SPDR Portfolio S&P 600 Sma...	U.S. Small Cap Equity	LONG
19	DON	0.0217	WisdomTree U.S. MidCap Dividend Fund	U.S. Mid Cap Dividend Eq...	LONG
26	IWN	0.0217	iShares Russell 2000 Value ETF	U.S. Small Cap Value	LONG
29	IJS	0.0217	iShares S&P SmallCap 600 Value ETF	U.S. Small Cap Value	LONG
32	SLYV	0.0217	State Street SPDR S&P 600 Small Cap Val...	U.S. Small Cap Value	LONG
34	VTWV	0.0217	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	LONG
47	SDVY	0.0217	First Trust SMID Cap Rising Dividend Ac...	Dividend Equity	LONG
7	FNX	0.0216	First Trust Mid Cap Core AlphaDEX Fund	U.S. Mid/Small Cap Equity	LONG
18	DES	0.0216	WisdomTree U.S. SmallCap Dividend Fund	U.S. Small Cap Dividend ...	LONG
20	EES	0.0216	WisdomTree U.S. SmallCap Fund	Small Cap Value Equities	LONG
33	VIOV	0.0216	Vanguard S&P Small-Cap 600 Value ETF	U.S. Small Cap Value	LONG
9	PRFZ	0.0215	Invesco RAFI US 1500 Small-Mid ETF	U.S. Small/Mid Cap Funda...	LONG
46	FYT	0.0215	First Trust Small Cap Value AlphaDEX Fu...	Value Equity	LONG
41	RWK	0.0214	Invesco S&P MidCap 400 Revenue ETF	U.S. Mid Cap Equity	LONG
38	RWJ	0.0213	Invesco S&P SmallCap 600 Revenue ETF	U.S. Small Cap Equity	LONG
48	ISMD	0.0213	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap Equity	LONG
39	RFV	0.0212	Invesco S&P MidCap 400 Pure Value ETF	Leveraged / Inverse	LONG
42	OMFS	0.0210	Invesco Russell 2000 Dynamic Multifacto...	U.S. Equity	LONG
13	FXO	0.0209	First Trust Financials AlphaDEX	Financials	FLAT
43	RZV	0.0209	Invesco S&P SmallCap 600 Pure Value ETF	Leveraged / Inverse	LONG
6	REGL	0.0207	ProShares S&P MidCap 400 Dividend Arist...	Dividend Equity	LONG
17	SMDV	0.0206	ProShares Russell 2000 Dividend Growers...	U.S. Dividend Equity	LONG
25	XSLV	0.0204	Invesco S&P SmallCap Low Volatility ETF	U.S. Small Cap Low Volat...	LONG
35	VFH	0.0204	Vanguard Financials ETF	Financials	FLAT
36	FNCL	0.0204	Fidelity MSCI Financials Index ETF	Financials	FLAT
27	SMMV	0.0203	iShares MSCI USA Small-Cap Min Vol Fact...	U.S. Small Cap Low Volat...	LONG
3	KBE	0.0202	State Street SPDR S&P Bank ETF	Financials	LONG
15	IYF	0.0202	iShares U.S. Financial ETF	Financials	FLAT
45	KCE	0.0202	State Street SPDR S&P Capital Markets E...	Financials	FLAT
16	IYG	0.0201	iShares U.S. Financial Services ETF	Financials	FLAT
24	XMLV	0.0198	Invesco S&P MidCap Low Volatility ETF	U.S. Mid Cap Low Volatil...	FLAT
1	XLF	0.0197	State Street Financial Select Sector SP...	Financials	FLAT
22	KBWB	0.0197	Invesco KBW Bank ETF	Financials	FLAT
2	KRE	0.0196	State Street SPDR S&P Regional Banking ...	Financials	LONG
14	IAT	0.0194	iShares U.S. Regional Banks ETF	Financials	LONG
44	IAI	0.0189	iShares U.S. Broker-Dealers & Securitie...	Financials	FLAT
40	PSP	0.0181	Invesco Global Listed Private Equity ETF	Global Equity	FLAT
4	KIE	0.0177	State Street SPDR S&P Insurance ETF	Financials	FLAT
5	REM	0.0173	iShares Mortgage Real Estate ETF	Real Estate	LONG

Cluster 8 – Emerging Markets Equity (19/25)**Members:** 25 **Var explained:** 0.875 **Signal:** LONG since 2026-04-10 (21 days, 9.9%)**Category breakdown**

category	n_members	abs_weight
Emerging Markets Equity	19	0.7724
Asia Equity	2	0.0812
Asia ex-Japan Equity	1	0.0418
Taiwan Equity	1	0.0357
Global Dividend Equity	1	0.0348
South Korea Equity	1	0.0341

Replication basket – IEMG+VWO+EEM (cor=0.99) basket lead at lag 1: -0.021

ticker	weight_ols	name	category	etf_signal
IEMG	0.776	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equi...	LONG
VWO	0.178	Vanguard FTSE Emerging Markets ETF	Emerging Markets Equi...	LONG
EEM	-0.012	iShares MSCI Emerging Index Fund	Emerging Markets Equi...	LONG

Members

	ticker	weight	name	category	etf_signal
21	IEMG	0.0425	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equity	LONG
2	EEM	0.0424	iShares MSCI Emerging Index Fund	Emerging Markets Equity	LONG
16	GEM	0.0423	Goldman Sachs ActiveBeta Emerging Marke...	Emerging Markets Equity	LONG
23	SPEM	0.0423	State Street SPDR Portfolio Emerging Ma...	Emerging Markets Equity	LONG
3	VWO	0.0422	Vanguard FTSE Emerging Markets ETF	Emerging Markets Equity	LONG
20	ESGE	0.0422	iShares ESG Aware MSCI EM ETF	Emerging Markets Equity	LONG
22	SCHE	0.0421	Schwab Emerging Markets Equity ETF	Emerging Markets Equity	LONG
15	EMGF	0.0419	iShares Emerging Markets Equity Factor ...	Emerging Markets Equity	LONG
18	XSOE	0.0419	WisdomTree Emerging Markets Ex-State Ow...	Emerging Markets Equity	LONG
5	AAXJ	0.0418	iShares MSCI All Country Asia ex Japan ...	Asia ex-Japan Equity	LONG
11	EEMA	0.0413	iShares MSCI Emerging Markets Asia ETF	Emerging Markets Equity	LONG
14	GMF	0.0411	State Street SPDR S&P Emerging Asia Pac...	Asia Equity	LONG
12	PXH	0.0410	Invesco RAFI Emerging Markets ETF	Emerging Markets Equity	LONG
17	FNDE	0.0410	Schwab Fundamental Emerging Markets Equ...	Emerging Markets Equity	LONG
19	EEMV	0.0404	iShares MSCI Emerging Markets Min Vol F...	Emerging Markets Equity	LONG
8	DGS	0.0402	WisdomTree Emerging Market SmallCap Fund	Emerging Markets Equity	LONG

10	AIA	0.0401	iShares Asia 50 ETF	Asia Equity	LONG
7	DEM	0.0396	WisdomTree Emerging Markets High Divide...	Emerging Markets Equity	LONG
6	EWX	0.0390	State Street SPDR S&P Emerging Markets ...	Emerging Markets Equity	LONG
4	FEM	0.0385	First Trust Emerging Markets AlphaDEX F...	Emerging Markets Equity	LONG
13	DVYE	0.0368	iShares Emerging Markets Dividend Index...	Emerging Markets Equity	FLAT
9	EWT	0.0357	iShares MSCI Taiwan ETF	Taiwan Equity	LONG
24	SDIV	0.0348	Global X SuperDividend ETF	Global Dividend Equity	LONG
25	FEMS	0.0347	First Trust Emerging Markets Small Cap ...	Emerging Markets Equity	FLAT
1	EWY	0.0341	iShares MSCI South Korea ETF	South Korea Equity	LONG

Cluster 9 – High Yield Bond (10) + Preferred Stock (7)

Members: 22 **Var explained:** 0.772 **Signal:** LONG since 2026-04-13 (20 days, 0.7%)

Category breakdown

category	n_members	abs_weight
High Yield Bond	10	0.4918
Preferred Stock	7	0.2893
Emerging Markets Fixed Income	4	0.1695
Unknown	1	0.0493

Replication basket – HYG+EMB+PFF (cor=0.99) basket lead at lag 1: 0.032

ticker	weight_ols	name	category	etf_signal
HYG	0.476	iShares iBoxx \$ High Yield Corporate ...	High Yield Bond	LONG
EMB	0.157	iShares J.P. Morgan USD Emerging Mark...	Emerging Markets Fixe...	FLAT
PFF	0.260	iShares Preferred and Income Securiti...	Preferred Stock	LONG

Members

	ticker	weight	name	category	etf_signal
14	JNK	0.0502	State Street SPDR Bloomberg High Yield ...	High Yield Bond	LONG
1	HYG	0.0501	iShares iBoxx \$ High Yield Corporate Bo...	High Yield Bond	LONG
20	HYLB	0.0500	Xtrackers USD High Yield Corporate Bond...	High Yield Bond	LONG
19	BBHY	0.0499	JPMorgan BetaBuilders USD High Yield Co...	High Yield Bond	LONG

17	SHYG	0.0497	iShares 0-5 Year High Yield Corporate B...	High Yield Bond	LONG
15	SJNK	0.0496	State Street SPDR Bloomberg Short Term ...	High Yield Bond	LONG
8	PHB	0.0493			LONG
16	ANGL	0.0492	VanEck Fallen Angel High Yield Bond ETF	High Yield Bond	LONG
18	FALN	0.0492	iShares Fallen Angels USD Bond ETF	High Yield Bond	LONG
6	HYS	0.0488	PIMCO 0-5 Year High Yield Corporat Bond...	High Yield Bond	LONG
10	PFF	0.0453	iShares Preferred and Income Securities...	Preferred Stock	LONG
3	HYLS	0.0450	First Trust Tactical High Yield ETF	High Yield Bond	LONG
2	EMB	0.0444	iShares J.P. Morgan USD Emerging Market...	Emerging Markets Fixed I...	FLAT
22	VWOB	0.0443	Vanguard Emerging Markets Government Bo...	Emerging Markets Fixed I...	FLAT
12	PFXF	0.0438	VanEck Preferred Securities ex Financia...	Preferred Stock	LONG
21	PCY	0.0435	Invesco Emerging Markets Sovereign Debt...	Emerging Markets Fixed I...	FLAT
7	PGX	0.0421	Invesco Preferred ETF	Preferred Stock	LONG
11	PSK	0.0416	State Street SPDR ICE Preferred Securit...	Preferred Stock	FLAT
4	FPE	0.0410	First Trust Preferred Securities and In...	Preferred Stock	LONG
5	PGF	0.0407	Invesco Financial Preferred ETF	Preferred Stock	FLAT
13	HYEM	0.0373	VanEck Emerging Markets High Yield Bond...	Emerging Markets Fixed I...	LONG
9	VRP	0.0348	Invesco Variable Rate Preferred ETF	Preferred Stock	LONG

Cluster 10 – U.S. Dividend Equity (11/20)

Members: 20 **Var explained:** 0.832 **Signal:** LONG since 2026-04-21 (14 days, 0.6%)

Category breakdown

category	n_members	abs_weight
U.S. Dividend Equity	11	0.5763
MLP / Energy Infrastructure	4	0.1667
Value Equity	2	0.1037
Dividend Equity	1	0.0524
U.S. Defensive Equity	1	0.0515
U.S. Low Volatility Equity	1	0.0495

Replication basket – SCHD+DVY+HDV (cor=0.98) basket lead at lag 1: 0.027

ticker	weight_ols	name	category	etf_signal
SCHD	0.131	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG
DVY	0.615	iShares Select Dividend ETF	U.S. Dividend Equity	LONG

HDV 0.236 iShares Core High Dividend ETF U.S. Dividend Equity LONG

Members

	ticker	weight	name	category	etf_signal
15	DVY	0.0538	iShares Select Dividend ETF	U.S. Dividend Equity	LONG
20	SPYD	0.0535	State Street SPDR Portfolio S&P 500 Hig...	U.S. Dividend Equity	LONG
14	SDOG	0.0534	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	LONG
6	DHS	0.0533	WisdomTree U.S. High Dividend Fund	U.S. Dividend Equity	LONG
19	ONEY	0.0528	State Street SPDR Russell 1000 Yield Fo...	U.S. Dividend Equity	LONG
11	FTA	0.0526	First Trust Large Cap Value AlphaDEX Fu...	Value Equity	FLAT
5	FDL	0.0524	First Trust Morningstar ETF	U.S. Dividend Equity	FLAT
18	SPHD	0.0524	Invesco S&P 500 High Dividend Low Volat...	Dividend Equity	LONG
10	SCHD	0.0521	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG
13	DIV	0.0517	Global X Super Dividend ETF	U.S. Dividend Equity	LONG
16	RDIV	0.0516	Invesco S&P Ultra Dividend Revenue ETF	U.S. Dividend Equity	LONG
7	CDC	0.0515	VictoryShares US EQ Income Enhanced Vol...	U.S. Defensive Equity	LONG
17	RPV	0.0511	Invesco S&P 500 Pure Value ETF	Value Equity	LONG
12	PEY	0.0510	Invesco High Yield Equity Dividend Achi...	U.S. Dividend Equity	LONG
9	HDV	0.0508	iShares Core High Dividend ETF	U.S. Dividend Equity	LONG
8	LVHD	0.0495	Franklin U.S. Low Volatility High Divid...	U.S. Low Volatility Equi...	FLAT
1	EMLP	0.0472	First Trust North American Energy Infra...	MLP / Energy Infrastruct...	LONG
4	MLPX	0.0419	Global X MLP & Energy Infrastructure ETF	MLP / Energy Infrastruct...	LONG
2	AMPLP	0.0393	Alerian MLP ETF	MLP / Energy Infrastruct...	LONG
3	MLPA	0.0383	Global X MLP ETF	MLP / Energy Infrastruct...	LONG

Cluster 11 – Healthcare (16/17)

Members: 17 **Var explained:** 0.764 **Signal:** LONG since 2026-05-06 (3 days, -2.0%)

Category breakdown

category	n_members	abs_weight
Healthcare	16	0.9419
U.S. Small Cap Healthcare	1	0.0581

Replication basket – XLV+VHT+XBI (cor=0.99) basket lead at lag 1: -0.007

ticker	weight_ols	name	category	etf_signal
XLV	-1.104	State Street Health Care Select Secto...	Healthcare	FLAT
VHT	1.989	Vanguard Health Care ETF	Healthcare	FLAT
XBI	0.128	State Street SPDR S&P Biotech ETF	Healthcare	FLAT

Members

	ticker	weight	name	category	etf_signal
15	VHT	0.0653	Vanguard Health Care ETF	Healthcare	FLAT
16	FHLC	0.0652	Fidelity MSCI Health Care Index ETF	Healthcare	FLAT
9	FXH	0.0645	First Trust Health Care AlphaDEX	Healthcare	FLAT
14	RSPH	0.0639	Invesco S&P 500 Equal Weight Health Car...	Healthcare	FLAT
13	IYH	0.0637	iShares U.S. Healthcare ETF	Healthcare	FLAT
10	IXJ	0.0622	iShares Global Healthcare ETF	Healthcare	FLAT
1	XLV	0.0620	State Street Health Care Select Sector ...	Healthcare	FLAT
3	IBB	0.0605	iShares Biotechnology ETF	Healthcare	LONG
6	FBT	0.0595	First Trust Amex Biotech Index Fund	Healthcare	FLAT
8	PSCH	0.0581	Invesco S&P SmallCap Health Care ETF	U.S. Small Cap Healthcare	LONG
7	XHE	0.0574	State Street SPDR S&P Health Care Equip...	Healthcare	FLAT
12	IHI	0.0573	iShares U.S. Medical Devices ETF	Healthcare	FLAT
5	PTH	0.0545	Invesco Dorsey Wright Healthcare Moment...	Healthcare	FLAT
17	PPH	0.0542	VanEck Pharmaceutical ETF	Healthcare	FLAT
4	XBI	0.0535	State Street SPDR S&P Biotech ETF	Healthcare	FLAT
2	ARKG	0.0511	ARK Genomic Revolution ETF	Healthcare	FLAT
11	IHF	0.0470	iShares U.S. Health Care Providers ETF	Healthcare	FLAT

Cluster 12 – Materials (13/17)**Members:** 17 **Var explained:** 0.756 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
Materials	13	0.7907
Consumer Staples	1	0.0589
Energy	1	0.0571
Norway Equity	1	0.0519

Industrial Metals	1	0.0414
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Replication basket – URA+GUNR+COPX (cor=0.97) basket lead at lag 1: 0.037

ticker	weight_ols	name	category	etf_signal
URA	0.069	Global X Uranium ETF	Materials	LONG
GUNR	0.681	FlexShares Global Upstream Natural Re...	Materials	FLAT
COPX	0.161	Global X Copper Miners ETF	Materials	LONG

Members

	ticker	weight	name	category	etf_signal
11	GNR	0.0653	State Street SPDR S&P Global Natural Re...	Materials	FLAT
6	MXI	0.0648	iShares Global Materials ETF	Materials	FLAT
10	GUNR	0.0643	FlexShares Global Upstream Natural Reso...	Materials	FLAT
7	IYM	0.0641	iShares U.S. Basic Materials ETF	Materials	FLAT
12	PICK	0.0627	iShares MSCI Global Select Metals & Min...	Materials	FLAT
9	VAW	0.0622	Vanguard Materials ETF	Materials	FLAT
17	NANR	0.0614	State Street SPDR S&P North American Na...	Materials	LONG
1	XLB	0.0612	State Street Materials Select Sector SP...	Materials	FLAT
8	RSPM	0.0611	Invesco S&P 500 Equal Weight Materials ...	Materials	FLAT
13	SLX	0.0606	VanEck Steel ETF	Materials	LONG
4	XME	0.0605	State Street SPDR S&P Metals & Mining E...	Materials	LONG
5	MOO	0.0589	VanEck Agribusiness ETF	Consumer Staples	LONG
3	COPX	0.0587	Global X Copper Miners ETF	Materials	LONG
16	IGE	0.0571	iShares North American Natural Resource...	Energy	LONG
14	NORW	0.0519	Global X MSCI Norway ETF	Norway Equity	LONG
2	URA	0.0438	Global X Uranium ETF	Materials	LONG
15	DBB	0.0414	Invesco DB Base Metals Fund	Industrial Metals	LONG

Cluster 13 – China Equity (13/15)

Members: 15 **Var explained:** 0.804 **Signal:** FLAT

Category breakdown

category	n_members	abs_weight
China Equity	13	0.8766
Emerging Markets Equity	1	0.0642
Hong Kong Equity	1	0.0591

Replication basket – KWEB+MCHI+FXI (cor=0.98) basket lead at lag 1: -0.040

ticker	weight_ols	name	category	etf_signal
KWEB	-0.038	KraneShares CSI China Internet ETF	China Equity	FLAT
MCHI	1.376	iShares MSCI China ETF	China Equity	FLAT
FXI	-0.395	iShares China Large-Cap ETF	China Equity	FLAT

Members

	ticker	weight	name	category	etf_signal
11	CXSE	0.0736	WisdomTree China ex-State-Owned Enterpr...	China Equity	FLAT
9	GXC	0.0735	State Street SPDR S&P China ETF	China Equity	FLAT
1	MCHI	0.0729	iShares MSCI China ETF	China Equity	FLAT
5	CQQQ	0.0712	Invesco China Technology ETF	China Equity	FLAT
2	FXI	0.0711	iShares China Large-Cap ETF	China Equity	FLAT
7	CHIQ	0.0706	Global X MSCI China Consumer Discretion...	China Equity	FLAT
4	KWEB	0.0679	KraneShares CSI China Internet ETF	China Equity	FLAT
6	ASHR	0.0669	Xtrackers Harvest CSI 300 China A-Share...	China Equity	LONG
8	CNYA	0.0661	iShares MSCI China A ETF	China Equity	LONG
13	KGRN	0.0651	KraneShares MSCI China Clean Technology...	China Equity	FLAT
3	EMQQ	0.0642	EMQQ The Emerging Markets Internet ETF	Emerging Markets Equity	FLAT
14	CNXT	0.0595	VanEck ChiNext Innovators ETF	China Equity	LONG
10	KBA	0.0594	KraneShares Boseri MSCI China A 50 Conn...	China Equity	LONG
12	EWI	0.0591	iShares MSCI Hong Kong Index Fund	Hong Kong Equity	LONG
15	KURE	0.0587	KraneShares MSCI All China Health Care ...	China Equity	FLAT

Cluster 14 – Real Estate (14/14)

Members: 14 **Var explained:** 0.962 **Signal:** LONG since 2026-04-10 (21 days, 4.2%)

Category breakdown

category	n_members	abs_weight
Real Estate	14	1

Replication basket – VNQ+SCHH+XLRE (cor=1.00) basket lead at lag 1: -0.001

ticker	weight_ols	name	category	etf_signal
VNQ	0.515	Vanguard Real Estate ETF	Real Estate	LONG
SCHH	0.537	Schwab U.S. REIT ETF	Real Estate	LONG
XLRE	-0.075	State Street Real Estate Select Secto...	Real Estate	LONG

Members

	ticker	weight	name	category	etf_signal
8	VNQ	0.0725	Vanguard Real Estate ETF	Real Estate	LONG
9	FREL	0.0725	Fidelity MSCI Real Estate Index ETF	Real Estate	LONG
11	SCHH	0.0725	Schwab U.S. REIT ETF	Real Estate	LONG
2	IYR	0.0724	iShares U.S. Real Estate ETF	Real Estate	LONG
6	RWR	0.0723	State Street SPDR Dow Jones REIT ETF	Real Estate	LONG
10	USRT	0.0723	iShares Core U.S. REIT ETF	Real Estate	LONG
13	PSR	0.0721	Invesco Active U.S. Real Estate Fund	Real Estate	LONG
1	XLRE	0.0719	State Street Real Estate Select Sector ...	Real Estate	LONG
5	ICF	0.0719	iShares Select U.S. REIT ETF	Real Estate	LONG
7	REET	0.0715	iShares Global REIT ETF	Real Estate	LONG
3	RWO	0.0711	State Street SPDR Dow Jones Global Real...	Real Estate	LONG
4	REZ	0.0698	iShares Residential and Multisector Rea...	Real Estate	LONG
14	NURE	0.0689	Nuveen Short-Term REIT ETF	Real Estate	LONG
12	INDS	0.0683	Pacer Industrial Real Estate ETF	Real Estate	LONG

Cluster 15 – Mixed: DBP, GLTR, GDXJ

Members: 13 **Var explained:** 0.831 **Signal:** FLAT

Category breakdown

category	n_members	abs_weight
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Materials	5	0.3889
Gold	3	0.2341
Precious Metals	2	0.1624
Silver	2	0.1525
Platinum	1	0.0622

Replication basket – GLD+IAU+SLV (cor=0.95) basket lead at lag 1: -0.017

ticker	weight_ols	name	category	etf_signal
GLD	0.115	SPDR Gold Shares	Gold	FLAT
IAU	0.623	iShares Gold Trust Shares	Gold	FLAT
SLV	0.418	iShares Silver Trust	Silver	LONG

Members

	ticker	weight	name	category	etf_signal
12	DBP	0.0814	Invesco DB Precious Metals Fund	Precious Metals	FLAT
7	GLTR	0.0810	abrdn Physical Precious Metals Basket S...	Precious Metals	FLAT
2	GDXJ	0.0786	VanEck Junior Gold Miners ETF	Materials	FLAT
1	GDX	0.0785	VanEck Gold Miners ETF	Materials	FLAT
3	GLD	0.0781	SPDR Gold Shares	Gold	FLAT
10	IAU	0.0781	iShares Gold Trust Shares	Gold	FLAT
11	SGOL	0.0779	abrdn Physical Gold Shares ETF	Gold	FLAT
13	GOAU	0.0777	US Global GO Gold and Precious Metal Mi...	Materials	FLAT
6	SIL	0.0776	Global X Silver Miners ETF	Materials	LONG
5	SILJ	0.0765	Amplify Junior Silver Miners ETF	Materials	LONG
4	SLV	0.0763	iShares Silver Trust	Silver	LONG
9	SIVR	0.0761	abrdn Physical Silver Shares ETF	Silver	LONG
8	PPLT	0.0622	abrdn Physical Platinum Shares ETF	Platinum	LONG

Cluster 16 – Municipal Bond (13/13)

Members: 13 **Var explained:** 0.819 **Signal:** FLAT

Category breakdown

category	n_members	abs_weight
Municipal Bond	13	1

Replication basket – MUB+VTEB+SUB (cor=0.97) basket lead at lag 1: 0.189

ticker	weight_ols	name	category	etf_signal
MUB	0.380	iShares National Muni Bond ETF	Municipal Bond	LONG
VTEB	0.468	Vanguard Tax-Exempt Bond ETF	Municipal Bond	FLAT
SUB	0.332	iShares Short-Term National Muni Bond...	Municipal Bond	LONG

Members

	ticker	weight	name	category	etf_signal
13	VTEB	0.0817	Vanguard Tax-Exempt Bond ETF	Municipal Bond	FLAT
11	MUB	0.0813	iShares National Muni Bond ETF	Municipal Bond	LONG
6	CMF	0.0801	iShares California Muni Bond ETF	Municipal Bond	LONG
9	TFI	0.0801	State Street SPDR Nuveen ICE Municipal ...	Municipal Bond	LONG
8	ITM	0.0800	VanEck Intermediate Muni ETF	Municipal Bond	LONG
1	FMB	0.0796	First Trust Managed Municipal ETF	Municipal Bond	LONG
7	NYF	0.0793	iShares New York Muni Bond ETF	Municipal Bond	LONG
5	PZA	0.0780	Invesco National AMT-Free Municipal Bon...	Municipal Bond	LONG
4	MUNI	0.0769	PIMCO Intermediate Municipal Bond Activ...	Municipal Bond	FLAT
2	HYD	0.0748	VanEck High Yield Muni ETF	Municipal Bond	LONG
3	HYMB	0.0720	State Street SPDR Nuveen ICE High Yield...	Municipal Bond	LONG
12	SUB	0.0694	iShares Short-Term National Muni Bond E...	Municipal Bond	LONG
10	SHM	0.0668	State Street SPDR Nuveen ICE Short Term...	Municipal Bond	FLAT

Cluster 17 – Consumer Staples (6/11)

Members: 11 **Var explained:** 0.748 **Signal:** LONG since 2026-05-01 (6 days, -2.3%)

Category breakdown

category	n_members	abs_weight
Consumer Staples	6	0.5726
Utilities	4	0.3528

U.S. Small Cap Consumer Staples 1 0.0746

Replication basket – XLU+XLP+VPU (cor=0.99) basket lead at lag 1: -0.011

ticker	weight_ols	name	category	etf_signal
XLU	-0.324	State Street Utilities Select Sector ...	Utilities	FLAT
XLP	0.584	State Street Consumer Staples Select ...	Consumer Staples	LONG
VPU	0.703	Vanguard Utilities ETF	Utilities	FLAT

Members

	ticker	weight	name	category	etf_signal
5	VDC	0.0983	Vanguard Consumer Staples ETF	Consumer Staples	LONG
6	FSTA	0.0983	Fidelity MSCI Consumer Staples Index ETF	Consumer Staples	LONG
1	XLP	0.0973	State Street Consumer Staples Select Se...	Consumer Staples	LONG
4	IYK	0.0947	iShares U.S. Consumer Staples ETF	Consumer Staples	LONG
3	KXI	0.0944	iShares Global Consumer Staples ETF	Consumer Staples	LONG
11	PBJ	0.0897	Invesco Food & Beverage ETF	Consumer Staples	FLAT
7	IDU	0.0892	iShares U.S. Utilities ETF	Utilities	FLAT
8	VPU	0.0880	Vanguard Utilities ETF	Utilities	FLAT
9	FUTY	0.0880	Fidelity MSCI Utilities Index ETF	Utilities	FLAT
2	XLU	0.0875	State Street Utilities Select Sector SP...	Utilities	FLAT
10	PSCC	0.0746	Invesco S&P SmallCap Consumer Staples E...	U.S. Small Cap Consumer ...	LONG

Cluster 18 – Mixed: ILF, EWZ, ECH

Members: 11 **Var explained:** 0.325 **Signal:** FLAT

Category breakdown

category	n_members	abs_weight
Latin America Equity	1	0.1565
Brazil Equity	1	0.1432
Chile Equity	1	0.1273
Malaysia Equity	1	0.1242
Indonesia Equity	1	0.1137

Philippines Equity	1	0.1051
Vietnam Equity	1	0.0792
Agricultural Commodities	1	0.0598
Turkey Equity	1	0.0575
Unknown	1	0.0189
Natural Gas	1	0.0147

Replication basket – EWZ+ILF+DBA+ECH (cor=0.93) basket lead at lag 1: 0.035

ticker	weight_ols	name	category	etf_signal
EWZ	0.047	iShares MSCI Brazil ETF	Brazil Equity	FLAT
ILF	0.382	iShares Latin America 40 ETF	Latin America Equity	FLAT
DBA	0.056	Invesco DB Agriculture Fund	Agricultural Commodit...	FLAT
ECH	0.158	iShares MSCI Chile ETF	Chile Equity	LONG

Members

	ticker	weight	name	category	etf_signal
4	ILF	0.1565	iShares Latin America 40 ETF	Latin America Equity	FLAT
1	EWZ	0.1432	iShares MSCI Brazil ETF	Brazil Equity	FLAT
3	ECH	0.1273	iShares MSCI Chile ETF	Chile Equity	LONG
8	EWM	0.1242	iShares MSCI Malaysia Index Fund	Malaysia Equity	LONG
6	EIDO	0.1137	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT
7	EPHE	0.1051	iShares MSCI Philippines ETF	Philippines Equity	FLAT
11	VNM	0.0792	VanEck Vietnam ETF	Vietnam Equity	FLAT
2	DBA	0.0598	Invesco DB Agriculture Fund	Agricultural Commodities	FLAT
10	TUR	0.0575	iShares MSCI Turkey ETF	Turkey Equity	LONG
9	RSX	0.0189			LONG
5	UNG	0.0147	United States Natural Gas Fund LP	Natural Gas	FLAT

Cluster 19 – Inflation-Protected Fixed Income (10/10)

Members: 10 **Var explained:** 0.908 **Signal:** LONG since 2026-04-09 (22 days, 0.8%)

Category breakdown

category	n_members	abs_weight
Inflation-Protected Fixed Income	10	1

Replication basket – VTIP+SCHP+STIP (cor=0.99) basket lead at lag 1: -0.013

ticker	weight_ols	name	category	etf_signal
VTIP	0.127	Vanguard Short-Term Inflation-Protect...	Inflation-Protected F...	FLAT
SCHP	0.748	Schwab U.S. TIPS ETF	Inflation-Protected F...	LONG
STIP	0.168	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected F...	FLAT

Members

	ticker	weight	name	category	etf_signal
4	TDTF	0.1033	FlexShares iBoxx 5 Year Target Duration...	Inflation-Protected Fixe...	LONG
6	TIPX	0.1032	State Street SPDR Bloomberg 1a??10 Year...	Inflation-Protected Fixe...	LONG
3	TIP	0.1028	iShares TIPS Bond ETF	Inflation-Protected Fixe...	LONG
8	SCHP	0.1023	Schwab U.S. TIPS ETF	Inflation-Protected Fixe...	LONG
5	TDTT	0.1007	FlexShares iBoxx 3 Year Target Duration...	Inflation-Protected Fixe...	LONG
9	STIP	0.0999	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
10	VTIP	0.0998	Vanguard Short-Term Inflation-Protected...	Inflation-Protected Fixe...	FLAT
7	SPIP	0.0997	State Street SPDR Portfolio TIPS ETF	Inflation-Protected Fixe...	LONG
2	STPZ	0.0995	PIMCO 1-5 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	LONG
1	LTPZ	0.0886	Pimco 15 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT

Cluster 20 – Japan Equity (7/9)

Members: 9 **Var explained:** 0.888 **Signal:** FLAT

Category breakdown

category	n_members	abs_weight
Japan Equity	7	0.7796
Asia-Pacific Equity	2	0.2204

Replication basket – EWJ+BBJP+VPL (cor=0.97) basket lead at lag 1: -0.039

ticker	weight_ols	name	category	etf_signal
EWJ	0.481	iShares MSCI Japan Index Fund	Japan Equity	LONG
BBJP	0.365	JPMorgan BetaBuilders Japan ETF	Japan Equity	LONG
VPL	0.072	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	LONG

Members

	ticker	weight	name	category	etf_signal
1	EWJ	0.1141	iShares MSCI Japan Index Fund	Japan Equity	LONG
8	BBJP	0.1141	JPMorgan BetaBuilders Japan ETF	Japan Equity	LONG
9	HEWJ	0.1115	iShares Currency Hedged MSCI Japan ETF	Japan Equity	LONG
6	DBJP	0.1114	Xtrackers MSCI Japan Hedged Equity ETF	Japan Equity	LONG
3	IPAC	0.1113	iShares Core MSCI Pacific ETF	Asia-Pacific Equity	FLAT
5	FLJH	0.1106	Franklin FTSE Japan Hedged ETF	Japan Equity	FLAT
2	DXJ	0.1097	WisdomTree Japan Hedged Equity Fund	Japan Equity	LONG
4	VPL	0.1091	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	LONG
7	EWJV	0.1082	iShares MSCI Japan Value ETF	Japan Equity	LONG

Cluster 21 – Broad Commodities (6/9)

Members: 9 **Var explained:** 0.913 **Signal:** LONG since 2026-01-08 (84 days, 47.8%)

Category breakdown

category	n_members	abs_weight
Broad Commodities	6	0.6698
Oil	2	0.2183
Energy Commodities	1	0.1119

Replication basket – PDBC+FTGC+DBC (cor=0.98) basket lead at lag 1: 0.072

ticker	weight_ols	name	category	etf_signal
PDBC	0.535	Invesco Optimum Yield Diversified Com...	Broad Commodities	LONG
FTGC	-0.029	First Trust Global Tactical Commodity...	Broad Commodities	FLAT

DBC	0.662	Invesco DB Commodity Index Tracking F...	Broad Commodities	LONG
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Members

	ticker	weight	name	category	etf_signal
5	GSG	0.1151	iShares GSCI Commodity-Indexed Trust Fu...	Broad Commodities	FLAT
2	DBC	0.1149	Invesco DB Commodity Index Tracking Fund	Broad Commodities	LONG
7	PDBC	0.1149	Invesco Optimum Yield Diversified Commo...	Broad Commodities	LONG
8	COMT	0.1148	iShares GSCI Commodity Dynamic Roll Str...	Broad Commodities	FLAT
9	DBE	0.1119	Invesco DB Energy Fund	Energy Commodities	FLAT
1	USO	0.1098	United States Oil Fund	Oil	FLAT
4	DBO	0.1085	Invesco DB Oil Fund	Oil	FLAT
6	DJP	0.1051	iPath Bloomberg Commodity Index Total R...	Broad Commodities	FLAT
3	FTGC	0.1050	First Trust Global Tactical Commodity S...	Broad Commodities	FLAT

Cluster 22 – Mixed: SRLN, BKLN, FTSL**Members:** 9 **Var explained:** 0.560 **Signal:** LONG since 2026-04-07 (24 days, 1.2%)**Category breakdown**

category	n_members	abs_weight
Bank Loan / Floating Rate	3	0.3632
Interest Rate Hedged Fixed Income	3	0.3245
Floating Rate Fixed Income	3	0.3123

Replication basket – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.048

ticker	weight_ols	name	category	etf_signal
FLOT	0.292	iShares Floating Rate Bond ETF	Floating Rate Fixed I...	FLAT
BKLN	0.209	Invesco Senior Loan ETF	Bank Loan / Floating ...	LONG
SRLN	0.311	State Street Blackstone Senior Loan E...	Bank Loan / Floating ...	LONG
FLRN	0.127	State Street SPDR Bloomberg Investmen...	Floating Rate Fixed I...	FLAT
FLTR	0.034	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	FLAT

Members

	ticker	weight	name	category	etf_signal
2	SRLN	0.1249	State Street Blackstone Senior Loan ETF	Bank Loan / Floating Rate	LONG
3	BKLN	0.1213	Invesco Senior Loan ETF	Bank Loan / Floating Rate	LONG
1	FTSL	0.1170	First Trust Senior Loan Fund	Bank Loan / Floating Rate	LONG
5	LQDH	0.1145	iShares Interest Rate Hedged Corporate ...	Interest Rate Hedged Fix...	LONG
7	FLOT	0.1081	iShares Floating Rate Bond ETF	Floating Rate Fixed Inco...	FLAT
6	IGBH	0.1057	iShares Interest Rate Hedged Long-Term ...	Interest Rate Hedged Fix...	LONG
4	IGHG	0.1043	ProShares Investment Grade-Interest Rat...	Interest Rate Hedged Fix...	LONG
8	FLRN	0.1035	State Street SPDR Bloomberg Investment ...	Floating Rate Fixed Inco...	FLAT
9	FLTR	0.1008	VanEck IG Floating Rate ETF	Floating Rate Fixed Inco...	FLAT

Cluster 23 – Cash / Ultra-Short Fixed Income (9/9)

Members: 9 **Var explained:** 0.457 **Signal:** FLAT

Category breakdown

category	n_members	abs_weight
Cash / Ultra-Short Fixed Income	9	1

Replication basket – BIL+SHV+MINT+GBIL+ICSH (cor=0.84) basket lead at lag 1: 0.106

	ticker	weight_ols	name	category	etf_signal
	BIL	-0.040	State Street SPDR Bloomberg 1-3 Month...	Cash / Ultra-Short Fi...	FLAT
	SHV	0.377	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fi...	FLAT
	MINT	0.176	PIMCO Enhanced Short Maturity Active ...	Cash / Ultra-Short Fi...	FLAT
	GBIL	0.118	Goldman Sachs Access Treasury 0-1 Yea...	Cash / Ultra-Short Fi...	FLAT
	ICSH	0.418	iShares Ultra Short Duration Bond Act...	Cash / Ultra-Short Fi...	FLAT

Members

	ticker	weight	name	category	etf_signal
5	GSY	0.1359	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fixed...	LONG
3	FTSM	0.1286	First Trust Enhanced Short Maturity ETF	Cash / Ultra-Short Fixed...	FLAT
7	SHV	0.1261	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fixed...	FLAT

6	ICSH	0.1244	iShares Ultra Short Duration Bond Activ...	Cash / Ultra-Short Fixed...	FLAT
4	NEAR	0.1090	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fixed...	FLAT
2	MINT	0.1062	PIMCO Enhanced Short Maturity Active Ex...	Cash / Ultra-Short Fixed...	FLAT
1	VNLA	0.1032	Janus Henderson Short Duration Income E...	Cash / Ultra-Short Fixed...	LONG
8	BIL	0.0879	State Street SPDR Bloomberg 1-3 Month T...	Cash / Ultra-Short Fixed...	FLAT
9	GBIL	0.0787	Goldman Sachs Access Treasury 0-1 Year ...	Cash / Ultra-Short Fixed...	FLAT

Cluster 24 – Currency (4/6)

Members: 6 **Var explained:** 0.742 **Signal:** FLAT

Category breakdown

category	n_members	abs_weight
Currency	4	0.641
International Fixed Income	2	0.359

Replication basket – BWX+IGOV+FXF (cor=0.98) basket lead at lag 1: 0.031

ticker	weight_ols	name	category	etf_signal
BWX	0.269	State Street SPDR Bloomberg Internati...	International Fixed I...	LONG
IGOV	0.315	iShares International Treasury Bond E...	International Fixed I...	LONG
FXF	0.306	Invesco CurrencyShares Swiss Franc Tr...	Currency	LONG

Members

	ticker	weight	name	category	etf_signal
2	IGOV	0.1801	iShares International Treasury Bond ETF	International Fixed Inco...	LONG
1	BWX	0.1789	State Street SPDR Bloomberg Internation...	International Fixed Inco...	LONG
4	FXE	0.1701	Invesco CurrencyShares Euro Currency Tr...	Currency	FLAT
5	FXF	0.1647	Invesco CurrencyShares Swiss Franc Trust	Currency	LONG
3	FXB	0.1609	Invesco CurrencyShares British Pound St...	Currency	LONG
6	FXY	0.1453	Invesco CurrencyShares Japanese Yen Tru...	Currency	FLAT

Cluster 25 – India Equity (3/4)**Members:** 4 **Var explained:** 0.964 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
India Equity	3	0.7508
Unknown	1	0.2492

Replication basket – INDА+INDY+EPI+PIN (cor=1.00) basket lead at lag 1: -0.071

ticker	weight_ols	name	category	etf_signal
INDA	0.250	iShares MSCI India ETF	India Equity	FLAT
INDY	0.246	iShares India 50 ETF	India Equity	FLAT
EPI	0.256	WisdomTree India Earnings Fund	India Equity	FLAT
PIN	0.249			FLAT

Members

	ticker	weight	name	category	etf_signal
1	INDA	0.2527	iShares MSCI India ETF	India Equity	FLAT
3	EPI	0.2493	WisdomTree India Earnings Fund	India Equity	FLAT
4	PIN	0.2492			FLAT
2	INDY	0.2489	iShares India 50 ETF	India Equity	FLAT

Cluster 26 – Currency (1/2)**Members:** 2 **Var explained:** 0.624 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
Currency	1	0.5
Leveraged / Inverse	1	0.5

Replication basket – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.017

ticker	weight_ols	name	category	etf_signal
SQQQ	0.507	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.496	Invesco DB USD Index Bullish Fund ETF	Currency	LONG

Members

ticker	weight	name	category	etf_signal
SQQQ	0.5	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.5	Invesco DB USD Index Bullish Fund ETF	Currency	LONG

Long ETF Positions (full)

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

ticker	name	category	cluster	cluster_name	cluster_signal	cluster_nss	entry_date	days	cum_return	prob_sell
GSY	Invesco Ultra Short Duration ...	Cash / Ultra-Shor...	cluster_23	Cash / Ultra-Short Fi...	FLAT	1.000	2022-11-11	874	20.4%	0.00
VNLA	Janus Henderson Short Duratio...	Cash / Ultra-Shor...	cluster_23	Cash / Ultra-Short Fi...	FLAT	1.000	2026-04-02	26	0.5%	0.03
SRLN	State Street Blackstone Senio...	Bank Loan / Float...	cluster_22	Mixed: SRLN, BKLN, FT...	LONG	1.000	2026-04-07	24	1.9%	0.00
FTSL	First Trust Senior Loan Fund	Bank Loan / Float...	cluster_22	Mixed: SRLN, BKLN, FT...	LONG	1.000	2026-04-06	25	1.4%	0.00
BKLN	Invesco Senior Loan ETF	Bank Loan / Float...	cluster_22	Mixed: SRLN, BKLN, FT...	LONG	1.000	2026-04-06	25	1.3%	0.00
LQDH	iShares Interest Rate Hedged ...	Interest Rate Hed...	cluster_22	Mixed: SRLN, BKLN, FT...	LONG	1.000	2026-04-06	25	1.3%	0.01
IGBH	iShares Interest Rate Hedged ...	Interest Rate Hed...	cluster_22	Mixed: SRLN, BKLN, FT...	LONG	1.000	2026-04-08	23	1.7%	0.01
IGHG	ProShares Investment Grade-In...	Interest Rate Hed...	cluster_22	Mixed: SRLN, BKLN, FT...	LONG	1.000	2026-04-07	24	1.1%	0.11
XSD	State Street SPDR S&P Semicon...	Technology	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-09	22	58.0%	0.00
NXTG	First Trust Indxx NextG ETF	Technology	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-08	23	21.7%	0.00
FTXL	First Trust Nasdaq Semiconduc...	Technology	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-07	24	59.1%	0.00
PTNQ	Pacer Trendpilot 100 ETF	Trend-Following E...	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-17	16	9.9%	0.00
PSI	Invesco Semiconductors ETF	Technology	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-07	24	56.9%	0.00
SOXX	iShares PHLX SOX Semiconducto...	Technology	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-08	23	43.8%	0.00
SMH	VanEck Semiconductor ETF	Technology	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-08	23	36.3%	0.00
IXN	iShares Global Tech ETF	Technology	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-10	21	24.5%	0.00
PSCT	Invesco S&P SmallCap Informat...	U.S. Small Cap Te...	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-07	24	30.0%	0.00
QQQ	Invesco QQQ Trust, Series 1	U.S. Large Cap Gr...	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-10	21	16.7%	0.00
QTUM	Defiance Quantum ETF	Technology	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-10	21	26.7%	0.00
PTLC	Pacer Trendpilot US Large Cap...	U.S. Trend-Follow...	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-20	15	4.3%	0.00
QYLD	Global X NASDAQ 100 Covered C...	Options Income	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-08	23	4.3%	0.00
PTF	Invesco Dorsey Wright Technol...	Technology	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-07	24	35.1%	0.00
BOTZ	Global X Robotics & Artificia...	Technology	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-16	17	12.5%	0.00
XT	iShares Future Exponential Te...	Technology	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-13	20	10.8%	0.00
TDIV	First Trust NASDAQ Technology...	Technology	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-13	20	14.3%	0.00
ROBO	ROBO Global Robotics and Auto...	Technology	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-14	19	14.2%	0.00
ONEQ	Fidelity Nasdaq Composite Ind...	U.S. Equity	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-13	20	13.4%	0.00
ICVT	iShares Convertible Bond ETF	Convertible Bond	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-08	23	10.4%	0.00
PWB	Invesco Large Cap Growth ETF	Growth Equity	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-10	21	13.7%	0.00
SPHB	Invesco S&P 500 High Beta ETF	U.S. Equity	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-10	21	16.4%	0.00
CWB	State Street SPDR Bloomberg C...	Convertible Bond	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-08	23	10.1%	0.00
IVW	iShares S&P 500 Growth ETF	U.S. Large Cap Gr...	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-13	20	11.5%	0.00
SPYG	State Street SPDR Portfolio S...	U.S. Large Cap Gr...	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-13	20	11.5%	0.00
VOOG	Vanguard S&P 500 Growth ETF	U.S. Large Cap Gr...	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-13	20	11.5%	0.00
IUSG	iShares Core S&P U.S. Growth ...	U.S. Large Cap Gr...	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-13	20	11.2%	0.00
IWY	iShares Russell Top 200 Growt...	Growth Equity	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-14	19	7.7%	0.01
VONG	Vanguard Russell 1000 Growth ...	U.S. Large Cap Gr...	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-14	19	7.3%	0.01

IWF	iShares Russell 1000 Growth F...	U.S. Large Cap Gr...	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-14	19	7.3%	0.02
ARKQ	ARK Autonomous Technology & R...	Technology	cluster_3	Technology (34/63)	FLAT	1.000	2026-04-14	19	11.1%	0.06
EWT	iShares MSCI Taiwan ETF	Taiwan Equity	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-08	23	27.2%	0.00
EWY	iShares MSCI South Korea ETF	South Korea Equity	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-09	22	38.5%	0.00
AIA	iShares Asia 50 ETF	Asia Equity	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-10	21	21.0%	0.00
AAXJ	iShares MSCI All Country Asia...	Asia ex-Japan Equ...	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-13	20	13.7%	0.00
EEMA	iShares MSCI Emerging Markets...	Emerging Markets ...	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-13	20	13.4%	0.00
EMGF	iShares Emerging Markets Equi...	Emerging Markets ...	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-10	21	13.3%	0.00
XSOE	WisdomTree Emerging Markets E...	Emerging Markets ...	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-10	21	12.5%	0.00
EEM	iShares MSCI Emerging Index F...	Emerging Markets ...	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-10	21	12.1%	0.00
IEMG	iShares Core MSCI Emerging Ma...	Emerging Markets ...	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-10	21	11.7%	0.00
ESGE	iShares ESG Aware MSCI EM ETF	Emerging Markets ...	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-13	20	11.2%	0.00
GEM	Goldman Sachs ActiveBeta Emer...	Emerging Markets ...	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-10	21	11.3%	0.00
GMF	State Street SPDR S&P Emergin...	Asia Equity	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-14	19	6.1%	0.00
DEM	WisdomTree Emerging Markets H...	Emerging Markets ...	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-09	22	7.0%	0.00
EEMV	iShares MSCI Emerging Markets...	Emerging Markets ...	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-10	21	8.5%	0.00
PXH	Invesco RAFI Emerging Markets...	Emerging Markets ...	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-10	21	5.7%	0.01
SPEM	State Street SPDR Portfolio E...	Emerging Markets ...	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-13	20	5.7%	0.02
VWO	Vanguard FTSE Emerging Market...	Emerging Markets ...	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-13	20	5.6%	0.02
SCHE	Schwab Emerging Markets Equit...	Emerging Markets ...	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-13	20	5.2%	0.03
EWX	State Street SPDR S&P Emergin...	Emerging Markets ...	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-10	21	8.5%	0.03
FNDE	Schwab Fundamental Emerging M...	Emerging Markets ...	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-10	21	4.9%	0.03
FEM	First Trust Emerging Markets ...	Emerging Markets ...	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-07	24	11.5%	0.09
DGS	WisdomTree Emerging Market Sm...	Emerging Markets ...	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-10	21	4.0%	0.15
SDIV	Global X SuperDividend ETF	Global Dividend E...	cluster_8	Emerging Markets Equi...	LONG	1.000	2026-04-09	22	1.2%	0.46
EIS	iShares MSCI Israel ETF	Israel Equity	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-10	21	8.4%	0.00
DSI	iShares ESG MSCI KLD 400 ETF	Large Cap Blend E...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-14	19	8.6%	0.00
IOO	iShares Global 100 ETF	Global Equity	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	9.1%	0.00
QAI	NYLI Hedge Multi-Strategy Tra...	Alternatives	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-08	23	4.0%	0.00
OEF	iShares S&P 100 Fund	U.S. Mega Cap Equ...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	9.5%	0.00
SUSA	iShares ESG Optimized MSCI US...	U.S. ESG Equity	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	8.0%	0.00
OMFL	Invesco Russell 1000 Dynamic ...	U.S. Multifactor ...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-10	21	7.2%	0.00
ESGU	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	8.1%	0.00
MGC	Vanguard Mega Cap ETF	U.S. Mega Cap Equ...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	8.7%	0.00
IVV	iShares Core S&P 500 ETF	U.S. Large Cap Eq...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.7%	0.00
SPY	State Street SPDR S&P 500 ETF...	U.S. Large Cap Eq...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.8%	0.00
VOO	Vanguard S&P 500 ETF	U.S. Large Cap Eq...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.7%	0.00
VV	Vanguard Large-Cap ETF	U.S. Large Cap Eq...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.7%	0.00
IWL	iShares Russell Top 200 ETF	U.S. Equity	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	8.2%	0.00
ILCB	iShares Morningstar Large-Cap...	U.S. Large Cap Eq...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.7%	0.00
SPTM	State Street SPDR Portfolio S...	U.S. Total Market...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.4%	0.00
IYY	iShares Dow Jones U.S. ETF	U.S. Total Market...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.5%	0.00
EPS	WisdomTree U.S. LargeCap Fund	Large Cap Value E...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-10	21	8.1%	0.00
SCHB	Schwab U.S. Broad Market ETF	U.S. Total Market...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.4%	0.00
ITOT	iShares Core S&P Total U.S. S...	U.S. Total Market...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.4%	0.00

SCHX	Schwab U.S. Large-Cap ETF	U.S. Large Cap Eq...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.5%	0.00
VTI	Vanguard Total Stock Market E...	U.S. Total Market...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.4%	0.00
SPYX	State Street SPDR S&P 500 Fos...	U.S. Equity	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.8%	0.00
IWB	iShares Russell 1000 ETF	U.S. Large Cap Eq...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.2%	0.00
SPHQ	Invesco S&P 500 Quality ETF	U.S. Quality / Fa...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-10	21	5.4%	0.00
IWV	iShares Russell 3000 Fund	U.S. Total Market...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.2%	0.00
VTHR	Vanguard Russell 3000 ETF	U.S. Total Market...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.3%	0.00
VONE	Vanguard Russell 1000 ETF	U.S. Large Cap Eq...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.2%	0.00
CATH	Global X S&P 500 Catholic Val...	Global Equity	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.3%	0.00
TILT	FlexShares Mornigstar US Mark...	U.S. Multifactor ...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	6.0%	0.00
ACWI	iShares MSCI ACWI ETF	Global Equity	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	6.2%	0.01
JHML	John Hancock Multifactor Larg...	U.S. Large Cap Eq...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	5.6%	0.01
SPGM	State Street SPDR Portfolio M...	Global Equity	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-10	21	7.0%	0.01
VT	Vanguard Total World Stock In...	Global Equity	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	6.1%	0.01
LRGF	iShares U.S. Equity Factor ETF	U.S. Multifactor ...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	6.4%	0.01
QDF	FlexShares Quality Dividend I...	Dividend Equity	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	5.3%	0.01
CRBN	iShares Low Carbon Optimized ...	Global Equity	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	6.2%	0.01
AOA	iShares Core 80/20 Aggressive...	Asset Allocation	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	4.8%	0.01
GSLC	Goldman Sachs ActiveBeta U.S....	Commodity	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	6.2%	0.01
URTH	iShares MSCI World ETF	Global Equity	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	5.7%	0.01
AOR	iShares Core 60/40 Balanced A...	Asset Allocation	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	3.6%	0.02
AOM	iShares Core 40/60 Moderate A...	Asset Allocation	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	2.4%	0.05
QUS	State Street SPDR MSCI USA St...	U.S. Quality / Fa...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-14	19	2.5%	0.06
FPXI	First Trust International Equ...	International IPO...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-10	21	7.3%	0.07
AOK	iShares Core 30/70 Conservati...	Asset Allocation	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	1.8%	0.07
QUAL	iShares MSCI USA Quality Fact...	U.S. Quality / Fa...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-14	19	3.6%	0.08
DGRW	WisdomTree U.S. Quality Divid...	U.S. Dividend Equ...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-14	19	3.2%	0.11
XLY	State Street Consumer Discret...	Consumer Discreti...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-15	18	1.0%	0.32
FDIS	Fidelity MSCI Consumer Discre...	Consumer Discreti...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-14	19	2.4%	0.34
VCR	Vanguard Consumer Discretion ...	Consumer Discreti...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-14	19	2.4%	0.35
VOX	Vanguard Communication Servic...	Communication Ser...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	2.3%	0.60
FCOM	Fidelity MSCI Communication S...	Communication Ser...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	2.1%	0.62
XLC	State Street Communication Se...	Communication Ser...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-14	19	-0.8%	0.74
IYC	iShares U.S. Consumer Discret...	Consumer Discreti...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-14	19	-1.0%	0.82
EWS	iShares MSCI Singapore ETF	Singapore Equity	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-07	24	1.9%	0.83
RTH	VanEck Retail ETF	Consumer Discreti...	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-10	21	-0.0%	0.93
SPLG			cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20		
VPL	Vanguard FTSE Pacific ETF	Asia-Pacific Equi...	cluster_20	Japan Equity (7/9)	FLAT	0.947	2026-04-10	21	10.1%	0.00
EWJ	iShares MSCI Japan Index Fund	Japan Equity	cluster_20	Japan Equity (7/9)	FLAT	0.947	2026-04-13	20	4.4%	0.03
BBJP	JPMorgan BetaBuilders Japan E...	Japan Equity	cluster_20	Japan Equity (7/9)	FLAT	0.947	2026-04-30	7	3.4%	0.03
HEWJ	iShares Currency Hedged MSCI ...	Japan Equity	cluster_20	Japan Equity (7/9)	FLAT	0.947	2026-04-08	23	2.7%	0.09
DBJP	Xtrackers MSCI Japan Hedged E...	Japan Equity	cluster_20	Japan Equity (7/9)	FLAT	0.947	2026-04-08	23	2.5%	0.13
EWJV	iShares MSCI Japan Value ETF	Japan Equity	cluster_20	Japan Equity (7/9)	FLAT	0.947	2026-05-06	3	-0.1%	0.17
DXJ	WisdomTree Japan Hedged Equit...	Japan Equity	cluster_20	Japan Equity (7/9)	FLAT	0.947	2026-05-01	6	4.2%	0.23
VLUE	iShares MSCI USA Value Factor...	U.S. Value Factor...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-09	22	21.2%	0.00

FNDX	Schwab Fundamental U.S. Large...	U.S. Fundamental ...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-09	22	5.6%	0.00
PRF	Invesco RAFI US 1000 ETF	U.S. Fundamental ...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-09	22	6.4%	0.00
IWX	iShares Russell Top 200 Value...	Value Equity	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	6.3%	0.00
NULV	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Va...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	6.7%	0.00
JVAL	JPMorgan U.S. Value Factor ETF	U.S. Value Factor...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	9.1%	0.00
IWD	iShares Russell 1000 Value ETF	U.S. Large Cap Va...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	5.8%	0.00
VONV	Vanguard Russell 1000 Value E...	U.S. Large Cap Va...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	5.8%	0.00
MGV	Vanguard Mega Cap Value ETF	U.S. Mega Cap Val...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-13	20	3.5%	0.00
ILCV	iShares Morningstar Large-Cap...	U.S. Large Cap Va...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-13	20	4.0%	0.00
VOOV	Vanguard S&P 500 Value ETF	U.S. Large Cap Va...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-14	19	3.3%	0.00
SPYV	State Street SPDR Portfolio S...	U.S. Large Cap Va...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-14	19	3.2%	0.00
IVE	iShares S&P 500 Value ETF	U.S. Large Cap Va...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-14	19	3.2%	0.00
SCHV	Schwab U.S. Large-Cap Value E...	U.S. Large Cap Va...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	4.4%	0.00
IUSV	iShares Core S&P U.S. Value E...	U.S. Large Cap Va...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-14	19	3.1%	0.00
RWL	Invesco S&P 500 Revenue ETF	U.S. Revenue Weig...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	4.8%	0.01
VTV	Vanguard Value ETF	U.S. Large Cap Va...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-13	20	2.8%	0.01
IMCB	iShares Morningstar Mid-Cap E...	U.S. Mid Cap Equi...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	5.5%	0.04
FEX	First Trust Large Cap Core Al...	Large Cap Blend E...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-09	22	4.2%	0.06
IWS	iShares Russell Mid-Cap Value...	U.S. Mid Cap Value	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-09	22	4.2%	0.06
IWR	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equi...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	4.9%	0.08
ETHO	Amplify Etho Climate Leadersh...	U.S. ESG Equity	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-09	22	5.6%	0.11
VO	Vanguard Mid-Cap ETF	U.S. Mid Cap Equi...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-13	20	3.3%	0.11
SIZE	iShares MSCI USA Size Factor ...	U.S. Size Factor ...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-13	20	3.4%	0.12
DTD	WisdomTree U.S. Total Dividen...	U.S. Dividend Equ...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-13	20	2.5%	0.12
AIVL	WisdomTree U.S. AI Enhanced V...	U.S. Value Factor...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	3.2%	0.12
DLN	WisdomTree U.S. LargeCap Divi...	U.S. Dividend Equ...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-13	20	2.6%	0.12
RSP	Invesco S&P 500 Equal Weight ...	U.S. Equal Weight...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-14	19	2.2%	0.20
VIG	Vanguard Div Appreciation ETF	U.S. Dividend Equ...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-14	19	2.0%	0.25
PFM	Invesco Dividend Achievers ETF	U.S. Dividend Equ...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-14	19	1.9%	0.26
FDVV	Fidelity High Dividend ETF	U.S. Dividend Equ...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-14	19	2.4%	0.27
DIA	State Street SPDR Dow Jones I...	U.S. Large Cap Eq...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-14	19	2.4%	0.29
JPME	JPMorgan Diversified Return U...	U.S. Equity	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-07	24	3.9%	0.34
VYM	Vanguard High Dividend Yield ...	U.S. Dividend Equ...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	2.5%	0.38
QUSA	ALPS O'Shares U.S. Quality Di...	Dividend Equity	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-20	15	-0.9%	0.47
DGRO	iShares Core Dividend Growth ...	U.S. Dividend Equ...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-13	20	1.2%	0.54
MOAT	VanEck Morningstar Wide Moat ...	U.S. Moat Equity	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-05-06	3	-0.9%	0.61
PWV	Invesco Large Cap Value ETF	Value Equity	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-02	26	3.6%	0.62
RDVY	First Trust Rising Dividend A...	U.S. Dividend Equ...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	3.0%	0.70
CFO	VictoryShares US 500 Enhanced...	U.S. Multifactor ...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-13	20	0.7%	0.72
SDY	State Street SPDR S&P Dividen...	U.S. Dividend Equ...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-05-01	6	-0.9%	0.71
CFA	VictoryShares US 500 Volatili...	U.S. Multifactor ...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-13	20	0.7%	0.73
USMV	iShares MSCI USA Min Vol Fact...	U.S. Low Volatili...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-05-05	4	-0.4%	0.64
EQAL	Invesco Russell 1000 Equal We...	U.S. Equal Weight...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-08	23	1.5%	0.76
ACWV	iShares MSCI Global Min Vol F...	Global Equity	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-05-06	3	-0.4%	0.66
ONEV	State Street SPDR Russell 100...	U.S. Equity	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-14	19	0.3%	0.85

JPUS	JPMorgan Diversified Return U...	U.S. Multifactor ...	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-08	23	0.9%	0.85
VOE	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-09	22	1.1%	0.85
PDBC	Invesco Optimum Yield Diversi...	Broad Commodities	cluster_21	Broad Commodities (6/...	LONG	0.918	2025-12-22	95	39.8%	0.01
DBC	Invesco DB Commodity Index Tr...	Broad Commodities	cluster_21	Broad Commodities (6/...	LONG	0.918	2025-12-22	95	38.7%	0.02
FXF	Invesco CurrencyShares Swiss ...	Currency	cluster_24	Currency (4/6)	FLAT	0.873	2026-05-05	4	0.6%	0.12
FXB	Invesco CurrencyShares Britis...	Currency	cluster_24	Currency (4/6)	FLAT	0.873	2026-04-13	20	1.0%	0.13
IGOV	iShares International Treasur...	International Fix...	cluster_24	Currency (4/6)	FLAT	0.873	2026-04-17	16	-0.1%	0.19
BWX	State Street SPDR Bloomberg I...	International Fix...	cluster_24	Currency (4/6)	FLAT	0.873	2026-05-06	3	-0.2%	0.35
CNYA	iShares MSCI China A ETF	China Equity	cluster_13	China Equity (13/15)	FLAT	0.858	2026-04-14	19	5.8%	0.00
ASHR	Xtrackers Harvest CSI 300 Chi...	China Equity	cluster_13	China Equity (13/15)	FLAT	0.858	2026-04-13	20	6.9%	0.01
EWK	iShares MSCI Hong Kong Index ...	Hong Kong Equity	cluster_13	China Equity (13/15)	FLAT	0.858	2026-04-08	23	2.3%	0.01
KBA	KraneShares Bosera MSCI China...	China Equity	cluster_13	China Equity (13/15)	FLAT	0.858	2026-04-10	21	9.8%	0.01
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_13	China Equity (13/15)	FLAT	0.858	2026-04-10	21	17.6%	0.01
FV	First Trust Dorsey Wright Foc...	Momentum Equity	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-14	19	8.9%	0.00
MTUM	iShares MSCI USA Momentum Fac...	U.S. Momentum Equ...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-08	23	18.1%	0.00
PRN	Invesco Dorsey Wright Industr...	Industrials	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-08	23	15.4%	0.00
FPX	First Trust US Equity Opportu...	IPO Equity	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-08	23	12.4%	0.00
RPX	Invesco S&P 500 Pure Growth E...	Growth Equity	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-09	22	14.2%	0.00
PDP	Invesco Dorsey Wright Momentu...	Momentum Equity	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-08	23	9.6%	0.00
VFMO	Vanguard U.S. Momentum Factor...	U.S. Momentum Equ...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-08	23	10.5%	0.00
XMMO	Invesco S&P MidCap Momentum E...	U.S. Mid Cap Mome...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-06	25	13.4%	0.00
SCHA	Schwab U.S. Small-Cap ETF	U.S. Small Cap Eq...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-09	22	8.6%	0.00
FTC	First Trust Large Cap Growth ...	Technology	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	9.2%	0.00
IMCG	iShares Morningstar Mid-Cap G...	U.S. Mid Cap Grow...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-09	22	9.1%	0.00
SCHM	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equi...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-09	22	7.2%	0.01
PSC	Principal U.S. Small-Cap ETF	U.S. Small Cap Eq...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	8.0%	0.01
IWM	iShares Russell 2000 Index Fu...	U.S. Small Cap Eq...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	9.2%	0.01
VTWO	Vanguard Russell 2000 ETF	U.S. Small Cap Eq...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	9.2%	0.01
IWO	iShares Russell 2000 Growth F...	U.S. Small Cap Gr...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-13	20	10.0%	0.01
VTWG	Vanguard Russell 2000 Growth ...	U.S. Small Cap Gr...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-13	20	10.0%	0.01
IWC	iShares Microcap ETF	U.S. Micro Cap Eq...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-09	22	10.0%	0.01
IJK	iShares S&P Mid-Cap 400 Growt...	U.S. Mid Cap Grow...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-09	22	6.9%	0.01
MDYG	State Street SPDR S&P 400 Mid...	U.S. Mid Cap Grow...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-09	22	6.9%	0.01
IVOG	Vanguard S&P Mid-Cap 400 Grow...	U.S. Mid Cap Grow...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-09	22	6.8%	0.02
VIOG	Vanguard S&P Small-Cap 600 Gr...	U.S. Small Cap Gr...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-08	23	6.6%	0.02
QMOM	Alpha Architect U.S. Quantita...	U.S. Momentum Equ...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-08	23	9.1%	0.02
SLYG	State Street SPDR S&P 600 Sma...	U.S. Small Cap Gr...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-08	23	6.6%	0.02
IJT	iShares S&P SmallCap 600 Grow...	U.S. Small Cap Gr...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-08	23	6.6%	0.02
PAVE	Global X U.S. Infrastructure ...	Industrials	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	4.4%	0.02
XSMO	Invesco S&P SmallCap Momentum...	U.S. Small Cap Mo...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-08	23	7.6%	0.02
VBK	Vanguard Small-Cap Growth ETF	U.S. Small Cap Gr...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	9.9%	0.03
XMHQ	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Qual...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	3.6%	0.03
ISCG	iShares Morningstar Small-Cap...	Growth Equity	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	7.4%	0.05
DWAS	Invesco Dorsey Wright SmallCa...	U.S. Small Cap Mo...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-07	24	13.7%	0.06
SMLF	iShares U.S. Small-Cap Equity...	U.S. Small Cap Mu...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-09	22	6.4%	0.06

VXF	Vanguard Extended Market ETF	U.S. Extended Mar...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	7.2%	0.07
IVOO	Vanguard S&P Mid-Cap 400 ETF		cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	4.8%	0.07
SPMD	State Street SPDR Portfolio S...	U.S. Mid Cap Equi...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	4.8%	0.07
MDY	State Street SPDR S&P MIDCAP ...	U.S. Mid Cap Equi...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	4.7%	0.07
IJH	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equi...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	4.7%	0.08
VB	Vanguard Small-Cap ETF	U.S. Small Cap Eq...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	6.1%	0.09
PSCI	Invesco S&P SmallCap Industri...	U.S. Small Cap In...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-13	20	2.7%	0.10
JETS	U.S. Global Jets ETF	Industrials	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-05-06	3	-2.3%	0.17
VIS	Vanguard Industrials ETF	Industrials	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-13	20	2.1%	0.24
JHMM	John Hancock Multifactor Mid ...	U.S. Mid Cap Equi...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	3.9%	0.24
FIDU	Fidelity MSCI Industrials Ind...	Industrials	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-13	20	2.1%	0.26
NUSC	Nuveen ESG Small-Cap ETF	U.S. Small Cap ES...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-13	20	3.0%	0.28
XLI	State Street Industrial Selec...	Industrials	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-13	20	1.3%	0.34
FTXR	First Trust Nasdaq Transporta...	Industrials	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-17	16	-0.5%	0.41
PKB	Invesco Building & Constructi...	Industrials	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-13	20	-0.2%	0.63
IYJ	iShares U.S. Industrials ETF	Industrials	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-13	20	0.7%	0.65
IYT	iShares U.S. Transportation E...	Industrials	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-14	19	1.4%	0.82
FXR	First Trust Industrials Alpha...	Industrials	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-14	19	-0.4%	0.85
XHB	State Street SPDR S&P Homebui...	Consumer Discreti...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-05-06	3	-3.9%	0.98
FXD	First Trust Cons. Discret. Al...	Consumer Discreti...	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-05-06	3	-3.2%	0.99
NURE	Nuveen Short-Term REIT ETF	Real Estate	cluster_14	Real Estate (14/14)	LONG	0.816	2026-04-13	20	5.7%	0.00
REZ	iShares Residential and Multi...	Real Estate	cluster_14	Real Estate (14/14)	LONG	0.816	2026-04-14	19	3.8%	0.01
USRT	iShares Core U.S. REIT ETF	Real Estate	cluster_14	Real Estate (14/14)	LONG	0.816	2026-04-09	22	5.2%	0.11
PSR	Invesco Active U.S. Real Esta...	Real Estate	cluster_14	Real Estate (14/14)	LONG	0.816	2026-04-09	22	4.9%	0.12
RWR	State Street SPDR Dow Jones R...	Real Estate	cluster_14	Real Estate (14/14)	LONG	0.816	2026-04-10	21	4.4%	0.16
SCHH	Schwab U.S. REIT ETF	Real Estate	cluster_14	Real Estate (14/14)	LONG	0.816	2026-04-09	22	4.4%	0.19
ICF	iShares Select U.S. REIT ETF	Real Estate	cluster_14	Real Estate (14/14)	LONG	0.816	2026-04-08	23	5.6%	0.19
FREL	Fidelity MSCI Real Estate Ind...	Real Estate	cluster_14	Real Estate (14/14)	LONG	0.816	2026-04-10	21	4.0%	0.26
XLRE	State Street Real Estate Sele...	Real Estate	cluster_14	Real Estate (14/14)	LONG	0.816	2026-04-09	22	4.3%	0.27
VNQ	Vanguard Real Estate ETF	Real Estate	cluster_14	Real Estate (14/14)	LONG	0.816	2026-04-10	21	4.0%	0.29
REET	iShares Global REIT ETF	Real Estate	cluster_14	Real Estate (14/14)	LONG	0.816	2026-04-13	20	3.4%	0.32
IYR	iShares U.S. Real Estate ETF	Real Estate	cluster_14	Real Estate (14/14)	LONG	0.816	2026-04-10	21	3.9%	0.32
RWO	State Street SPDR Dow Jones G...	Real Estate	cluster_14	Real Estate (14/14)	LONG	0.816	2026-04-13	20	3.0%	0.34
INDS	Pacer Industrial Real Estate ...	Real Estate	cluster_14	Real Estate (14/14)	LONG	0.816	2026-04-14	19	2.0%	0.51
SCHD	Schwab US Dividend Equity ETF	U.S. Dividend Equ...	cluster_10	U.S. Dividend Equity ...	LONG	0.787	2026-04-21	14	2.5%	0.00
RDIV	Invesco S&P Ultra Dividend Re...	U.S. Dividend Equ...	cluster_10	U.S. Dividend Equity ...	LONG	0.787	2026-04-21	14	-0.2%	0.01
SPYD	State Street SPDR Portfolio S...	U.S. Dividend Equ...	cluster_10	U.S. Dividend Equity ...	LONG	0.787	2026-04-21	14	1.0%	0.01
AML	Alerian MLP ETF	MLP / Energy Infr...	cluster_10	U.S. Dividend Equity ...	LONG	0.787	2026-04-29	8	0.6%	0.01
RPV	Invesco S&P 500 Pure Value ETF	Value Equity	cluster_10	U.S. Dividend Equity ...	LONG	0.787	2026-04-20	15	0.3%	0.08
DIV	Global X Super Dividend ETF	U.S. Dividend Equ...	cluster_10	U.S. Dividend Equity ...	LONG	0.787	2026-04-07	24	1.3%	0.12
MLPA	Global X MLP ETF	MLP / Energy Infr...	cluster_10	U.S. Dividend Equity ...	LONG	0.787	2026-04-29	8	-1.5%	0.03
HDV	iShares Core High Dividend ETF	U.S. Dividend Equ...	cluster_10	U.S. Dividend Equity ...	LONG	0.787	2026-05-01	6	-1.0%	0.14
SDOG	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equ...	cluster_10	U.S. Dividend Equity ...	LONG	0.787	2026-04-07	24	1.7%	0.28
DHS	WisdomTree U.S. High Dividend...	U.S. Dividend Equ...	cluster_10	U.S. Dividend Equity ...	LONG	0.787	2026-04-20	15	-0.1%	0.30
PEY	Invesco High Yield Equity Div...	U.S. Dividend Equ...	cluster_10	U.S. Dividend Equity ...	LONG	0.787	2026-04-13	20	2.3%	0.33

SPHD	Invesco S&P 500 High Dividend...	Dividend Equity	cluster_10	U.S. Dividend Equity ...	LONG	0.787	2026-05-05	4	-0.5%	0.30
MLPX	Global X MLP & Energy Infrast...	MLP / Energy Infr...	cluster_10	U.S. Dividend Equity ...	LONG	0.787	2026-04-30	7	-3.5%	0.07
ONEY	State Street SPDR Russell 100...	U.S. Dividend Equ...	cluster_10	U.S. Dividend Equity ...	LONG	0.787	2026-04-10	21	2.1%	0.52
DVY	iShares Select Dividend ETF	U.S. Dividend Equ...	cluster_10	U.S. Dividend Equity ...	LONG	0.787	2026-04-08	23	-0.6%	0.82
CDC	VictoryShares US EQ Income En...	U.S. Defensive Eq...	cluster_10	U.S. Dividend Equity ...	LONG	0.787	2026-04-09	22	-0.6%	0.87
EMLP	First Trust North American En...	MLP / Energy Infr...	cluster_10	U.S. Dividend Equity ...	LONG	0.787	2026-04-28	9	-0.6%	0.56
TDTT	FlexShares iBoxx 3 Year Targe...	Inflation-Protect...	cluster_19	Inflation-Protected F...	LONG	0.787	2026-04-01	27	1.2%	0.12
STPZ	PIMCO 1-5 Year U.S. TIPS Inde...	Inflation-Protect...	cluster_19	Inflation-Protected F...	LONG	0.787	2026-04-02	26	0.9%	0.14
TDTF	FlexShares iBoxx 5 Year Targe...	Inflation-Protect...	cluster_19	Inflation-Protected F...	LONG	0.787	2026-04-09	22	1.0%	0.19
TIPX	State Street SPDR Bloomberg 1...	Inflation-Protect...	cluster_19	Inflation-Protected F...	LONG	0.787	2026-04-07	24	0.8%	0.21
TIP	iShares TIPS Bond ETF	Inflation-Protect...	cluster_19	Inflation-Protected F...	LONG	0.787	2026-04-10	21	0.8%	0.22
SCHP	Schwab U.S. TIPS ETF	Inflation-Protect...	cluster_19	Inflation-Protected F...	LONG	0.787	2026-04-13	20	0.6%	0.23
SPIP	State Street SPDR Portfolio T...	Inflation-Protect...	cluster_19	Inflation-Protected F...	LONG	0.787	2026-04-10	21	0.7%	0.27
ENZL	iShares MSCI New Zealand ETF	New Zealand Equity	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-05-06	3	0.4%	0.01
VIDI	Vident International Equity S...	International Equ...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-09	22	6.1%	0.02
FNDF	Schwab Fundamental Internatio...	International Equ...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-08	23	4.2%	0.02
IQDF	FlexShares International Qual...	International Equ...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	5.0%	0.03
PXF	Invesco RAFI Developed Market...	International Equ...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-08	23	4.2%	0.03
GWX	State Street SPDR S&P Interna...	International Equ...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-13	20	5.1%	0.03
VEU	Vanguard FTSE All World Ex US...	Global ex-US Equi...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	5.0%	0.04
HFXI	NYLI FTSE International Equit...	International Equ...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-09	22	4.7%	0.05
VXUS	Vanguard Total International ...	Global ex-US Equi...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	5.1%	0.05
IXUS	iShares Core MSCI Total Inter...	Global ex-US Equi...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	5.1%	0.06
CWI	State Street SPDR MSCI ACWI e...	Global ex-US Equi...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	4.6%	0.06
SPDW	State Street SPDR Portfolio D...	Developed ex-US E...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	4.6%	0.07
SCHF	Schwab International Equity E...	Developed ex-US E...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	4.6%	0.07
ACWX	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equi...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	4.7%	0.07
VEA	Vanguard FTSE Developed Marke...	Developed ex-US E...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	4.5%	0.08
VSS	Vanguard FTSE All-Wld ex-US S...	Global ex-US Smal...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-13	20	4.8%	0.08
FXA	Invesco CurrencyShares Austra...	Currency	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-13	20	2.1%	0.10
EWN	iShares MSCI Netherlands Inde...	Netherlands Equity	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	4.9%	0.11
IPKW	Invesco International BuyBack...	International Equ...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-09	22	2.5%	0.13
FNDC	Schwab Fundamental Internatio...	International Equ...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	3.7%	0.13
SCZ	iShares MSCI EAFE Small-Cap E...	Developed ex-US E...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	3.7%	0.16
SCHC	Schwab International Small-Ca...	Developed ex-US E...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-13	20	3.4%	0.20
DLS	WisdomTree International Smal...	International Equ...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-13	20	2.2%	0.20
IDV	iShares International Select ...	International Div...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-06	25	4.4%	0.26
FXC	Invesco CurrencyShares Canadi...	Currency	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-20	15	-0.2%	0.27
FEP	First Trust Europe AlphaDEX F...	Europe Equity	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-09	22	2.7%	0.28
EWI	iShares MSCI Italy ETF	Italy Equity	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-08	23	4.7%	0.28
EWX	iShares MSCI Mexico ETF	Mexico Equity	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-05-05	4	3.4%	0.33
VNQI	Vanguard Global ex-U.S. Real ...	Real Estate	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-05-04	5	2.6%	0.41
INTM	iShares MSCI Intl Momentum Fa...	International Equ...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	2.7%	0.41
EWG	iShares MSCI Germany Index Fu...	Germany Equity	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-14	19	0.8%	0.41
EWO	iShares MSCI Austria ETF	Austria Equity	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	4.2%	0.43

VIGI	Vanguard International Divide...	International Div...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-14	19	0.5%	0.43
NFRA	FlexShares STOXX Global Broad...	Global Equity	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-09	22	-0.0%	0.46
VYMI	Vanguard International High D...	International Div...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-09	22	1.7%	0.46
BBAX	JPMorgan BetaBuilders Develop...	Asia-Pacific Equi...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-30	7	0.9%	0.51
DWX	State Street SPDR S&P Interna...	International Div...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-05-04	5	1.0%	0.54
EWL	iShares MSCI Switzerland ETF	Switzerland Equity	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-15	18	-0.2%	0.57
PID	Invesco International Dividen...	International Div...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-30	7	-0.0%	0.58
IHDG	WisdomTree International Hedg...	Alternatives	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-05-06	3	-1.5%	0.69
RWX	State Street SPDR Dow Jones I...	Real Estate	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-05-06	3	-1.4%	0.77
IDLV	Invesco S&P International Dev...	International Equ...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-30	7	-0.6%	0.78
EWG	iShares MSCI Canada Index Fund	Canada Equity	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-09	22	3.0%	0.81
EZA	iShares MSCI South Africa Ind...	South Africa Equi...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-05-08	1	0.1%	0.67
EBND	State Street SPDR Bloomberg E...	Emerging Markets ...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-05-06	3	0.0%	0.81
EMLC	VanEck J. P. Morgan EM Local ...	Emerging Markets ...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-05-06	3	-0.0%	0.84
HDEF	Xtrackers MSCI EAFE High Divi...	International Equ...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-07	24	0.5%	0.93
EWD	iShares MSCI Sweden ETF	Sweden Equity	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-05-06	3	-2.6%	0.96
EWU	iShares MSCI United Kingdom E...	United Kingdom Eq...	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-30	7	-1.1%	0.98
IGF	iShares Global Infrastructure...	Global Equity	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-30	7	-1.6%	0.76
SLX	VanEck Steel ETF	Materials	cluster_12	Materials (13/17)	FLAT	0.550	2026-04-09	22	10.8%	0.00
XME	State Street SPDR S&P Metals ...	Materials	cluster_12	Materials (13/17)	FLAT	0.550	2026-04-15	18	7.7%	0.04
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_12	Materials (13/17)	FLAT	0.550	2026-04-10	21	6.8%	0.25
URA	Global X Uranium ETF	Materials	cluster_12	Materials (13/17)	FLAT	0.550	2026-04-14	19	8.3%	0.35
COPX	Global X Copper Miners ETF	Materials	cluster_12	Materials (13/17)	FLAT	0.550	2026-05-08	1	1.9%	0.55
NORW	Global X MSCI Norway ETF	Norway Equity	cluster_12	Materials (13/17)	FLAT	0.550	2025-12-09	104	31.8%	0.63
NANR	State Street SPDR S&P North A...	Materials	cluster_12	Materials (13/17)	FLAT	0.550	2026-05-11	0	0.0%	0.70
IGE	iShares North American Natura...	Energy	cluster_12	Materials (13/17)	FLAT	0.550	2026-05-01	6	0.1%	0.64
MOO	VanEck Agribusiness ETF	Consumer Staples	cluster_12	Materials (13/17)	FLAT	0.550	2026-05-06	3	-2.8%	0.86
PFXF	VanEck Preferred Securities e...	Preferred Stock	cluster_9	High Yield Bond (10) ...	LONG	0.489	2026-04-14	19	4.1%	0.00
PFF	iShares Preferred and Income ...	Preferred Stock	cluster_9	High Yield Bond (10) ...	LONG	0.489	2026-04-15	18	1.5%	0.05
HYEM	VanEck Emerging Markets High ...	Emerging Markets ...	cluster_9	High Yield Bond (10) ...	LONG	0.489	2026-04-13	20	1.3%	0.12
VRP	Invesco Variable Rate Preferr...	Preferred Stock	cluster_9	High Yield Bond (10) ...	LONG	0.489	2026-04-10	21	0.5%	0.45
FPE	First Trust Preferred Securit...	Preferred Stock	cluster_9	High Yield Bond (10) ...	LONG	0.489	2026-04-14	19	0.6%	0.46
PGX	Invesco Preferred ETF	Preferred Stock	cluster_9	High Yield Bond (10) ...	LONG	0.489	2026-04-17	16	-0.4%	0.58
HYLS	First Trust Tactical High Yie...	High Yield Bond	cluster_9	High Yield Bond (10) ...	LONG	0.489	2026-04-09	22	0.5%	0.64
SHYG	iShares 0-5 Year High Yield C...	High Yield Bond	cluster_9	High Yield Bond (10) ...	LONG	0.489	2026-04-07	24	0.9%	0.76
FALN	iShares Fallen Angels USD Bon...	High Yield Bond	cluster_9	High Yield Bond (10) ...	LONG	0.489	2026-04-10	21	0.6%	0.84
JNK	State Street SPDR Bloomberg H...	High Yield Bond	cluster_9	High Yield Bond (10) ...	LONG	0.489	2026-04-08	23	0.4%	0.87
SJNK	State Street SPDR Bloomberg S...	High Yield Bond	cluster_9	High Yield Bond (10) ...	LONG	0.489	2026-04-07	24	0.9%	0.89
HYS	PIMCO 0-5 Year High Yield Cor...	High Yield Bond	cluster_9	High Yield Bond (10) ...	LONG	0.489	2026-04-08	23	0.1%	0.89
BBHY	JPMorgan BetaBuilders USD Hig...	High Yield Bond	cluster_9	High Yield Bond (10) ...	LONG	0.489	2026-04-08	23	0.3%	0.90
HYG	iShares iBoxx \$ High Yield Co...	High Yield Bond	cluster_9	High Yield Bond (10) ...	LONG	0.489	2026-04-08	23	0.3%	0.91
HYLB	Xtrackers USD High Yield Corp...	High Yield Bond	cluster_9	High Yield Bond (10) ...	LONG	0.489	2026-04-08	23	0.2%	0.92
ANGL	VanEck Fallen Angel High Yiel...	High Yield Bond	cluster_9	High Yield Bond (10) ...	LONG	0.489	2026-04-09	22	0.2%	0.95
PHB			cluster_9	High Yield Bond (10) ...	LONG	0.489	2026-04-10	21		
IWN	iShares Russell 2000 Value ETF	U.S. Small Cap Va...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	5.8%	0.01

VTWV	Vanguard Russell 2000 Value E...	U.S. Small Cap Va...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	5.7%	0.01
OMFS	Invesco Russell 2000 Dynamic ...	U.S. Equity	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	6.3%	0.02
ISMD	Inspire Small/Mid Cap ETF	U.S. Small/Mid Ca...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	7.5%	0.02
EES	WisdomTree U.S. SmallCap Fund	Small Cap Value E...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-08	23	5.1%	0.02
VIOO	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Eq...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	5.0%	0.04
IJR	iShares Core S&P Small-Cap ETF	U.S. Small Cap Eq...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	5.0%	0.05
SPSM	State Street SPDR Portfolio S...	U.S. Small Cap Eq...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	5.0%	0.05
PRFZ	Invesco RAFI US 1500 Small-Mi...	U.S. Small/Mid Ca...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	6.8%	0.05
FNDA	Schwab Fundamental U.S. Small...	U.S. Small Cap Fu...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	5.4%	0.09
FYX	First Trust Small Cap Core Al...	U.S. Small Cap Eq...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-08	23	6.5%	0.11
FNX	First Trust Mid Cap Core Alph...	U.S. Mid/Small Ca...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	3.9%	0.12
RWJ	Invesco S&P SmallCap 600 Reve...	U.S. Small Cap Eq...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	4.6%	0.15
SLYV	State Street SPDR S&P 600 Sma...	U.S. Small Cap Va...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	4.5%	0.16
IJS	iShares S&P SmallCap 600 Valu...	U.S. Small Cap Va...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	4.5%	0.17
RFV	Invesco S&P MidCap 400 Pure V...	Leveraged / Inver...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	2.9%	0.17
VIOV	Vanguard S&P Small-Cap 600 Va...	U.S. Small Cap Va...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	4.5%	0.18
EZM	WisdomTree U.S. MidCap Fund	Mid Cap Value Equ...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	3.7%	0.23
VBR	Vanguard Small-Cap Value ETF	U.S. Small Cap Va...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	3.3%	0.32
DES	WisdomTree U.S. SmallCap Divi...	U.S. Small Cap Di...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-08	23	3.1%	0.39
RWK	Invesco S&P MidCap 400 Revenu...	U.S. Mid Cap Equi...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	2.7%	0.51
RZV	Invesco S&P SmallCap 600 Pure...	Leveraged / Inver...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-13	20	3.5%	0.58
IJJ	iShares S&P Mid-Cap 400 Value...	U.S. Mid Cap Value	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	2.2%	0.65
IVOV	Vanguard S&P Mid-Cap 400 Valu...	U.S. Mid Cap Value	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	2.1%	0.65
MDYV	State Street SPDR S&P 400 Mid...	U.S. Mid Cap Value	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	2.2%	0.65
DON	WisdomTree U.S. MidCap Divide...	U.S. Mid Cap Divi...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-13	20	0.9%	0.72
FYT	First Trust Small Cap Value A...	Value Equity	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-07	24	4.3%	0.82
XSLV	Invesco S&P SmallCap Low Vola...	U.S. Small Cap Lo...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	-0.6%	0.93
SDVY	First Trust SMID Cap Rising D...	Dividend Equity	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	1.0%	0.93
KRE	State Street SPDR S&P Regiona...	Financials	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	-1.9%	0.93
SMDV	ProShares Russell 2000 Divide...	U.S. Dividend Equ...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	-1.0%	0.94
REM	iShares Mortgage Real Estate ...	Real Estate	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-13	20	0.1%	0.95
KBE	State Street SPDR S&P Bank ETF	Financials	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	-1.1%	0.97
REGL	ProShares S&P MidCap 400 Divi...	Dividend Equity	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	-0.9%	1.00
IAT	iShares U.S. Regional Banks E...	Financials	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	-1.4%	1.00
SMMV	iShares MSCI USA Small-Cap Mi...	U.S. Small Cap Lo...	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	-1.2%	1.00
HYD	VanEck High Yield Muni ETF	Municipal Bond	cluster_16	Municipal Bond (13/13)	FLAT	0.244	2026-04-08	23	0.9%	0.50
HYMB	State Street SPDR Nuveen ICE ...	Municipal Bond	cluster_16	Municipal Bond (13/13)	FLAT	0.244	2026-04-09	22	0.5%	0.53
FMB	First Trust Managed Municipal...	Municipal Bond	cluster_16	Municipal Bond (13/13)	FLAT	0.244	2026-04-17	16	0.0%	0.67
NYF	iShares New York Muni Bond ETF	Municipal Bond	cluster_16	Municipal Bond (13/13)	FLAT	0.244	2026-04-15	18	0.4%	0.69
PZA	Invesco National AMT-Free Mun...	Municipal Bond	cluster_16	Municipal Bond (13/13)	FLAT	0.244	2026-04-10	21	0.6%	0.73
ITM	VanEck Intermediate Muni ETF	Municipal Bond	cluster_16	Municipal Bond (13/13)	FLAT	0.244	2026-05-08	1	0.0%	0.75
MUB	iShares National Muni Bond ETF	Municipal Bond	cluster_16	Municipal Bond (13/13)	FLAT	0.244	2026-04-17	16	-0.1%	0.77
CMF	iShares California Muni Bond ...	Municipal Bond	cluster_16	Municipal Bond (13/13)	FLAT	0.244	2026-05-06	3	0.0%	0.86
TFI	State Street SPDR Nuveen ICE ...	Municipal Bond	cluster_16	Municipal Bond (13/13)	FLAT	0.244	2026-05-06	3	-0.0%	0.89
SUB	iShares Short-Term National M...	Municipal Bond	cluster_16	Municipal Bond (13/13)	FLAT	0.244	2026-05-11	0	0.0%	0.90

EWM	iShares MSCI Malaysia Index F...	Malaysia Equity	cluster_18	Mixed: ILF, EWZ, ECH	FLAT	0.072	2026-04-22	13	3.3%	0.00
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_18	Mixed: ILF, EWZ, ECH	FLAT	0.072	2026-04-10	21	3.9%	0.31
ECH	iShares MSCI Chile ETF	Chile Equity	cluster_18	Mixed: ILF, EWZ, ECH	FLAT	0.072	2026-05-06	3	-2.0%	0.98
RSX			cluster_18	Mixed: ILF, EWZ, ECH	FLAT	0.072	2023-07-27	699		
PSCH	Invesco S&P SmallCap Health C...	U.S. Small Cap He...	cluster_11	Healthcare (16/17)	LONG	0.021	2026-04-13	20	4.8%	0.06
IBB	iShares Biotechnology ETF	Healthcare	cluster_11	Healthcare (16/17)	LONG	0.021	2026-05-06	3	-2.2%	0.81
LDUR	PIMCO Enhanced Low Duration A...	Fixed Income	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-04-07	24	0.4%	0.29
CMBS	iShares CMBS Bond ETF	Mortgage-Backed F...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.1%	0.41
IAGG	iShares International Aggrega...	International Fix...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.3%	0.42
BNDX	Vanguard Total International ...	International Fix...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.3%	0.41
SPSB	State Street SPDR Portfolio S...	Fixed Income	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-04-09	22	0.1%	0.71
IGLB	iShares 10 Year Investment Gr...	Investment Grade ...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.4%	0.76
USIG	iShares Broad USD Investment ...	Investment Grade ...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-06	3	-0.2%	0.81
VCLT	Vanguard Long-Term Corporate ...	Investment Grade ...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.4%	0.78
GTO	Invesco Total Return Bond ETF	Active Fixed Inco...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-07	2	0.0%	0.80
FLCO	Franklin Investment Grade Cor...	Active Fixed Inco...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-07	2	0.3%	0.81
SPLB	State Street SPDR Portfolio L...	Investment Grade ...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.3%	0.78
CORP	PIMCO Investment Grade Corpor...	Investment Grade ...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-06	3	-0.2%	0.83
GNMA	iShares GNMA Bond ETF	Mortgage-Backed F...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-07	2	-0.1%	0.79
BAB	Invesco Taxable Municipal Bon...	Taxable Municipal...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.65
IGIB	iShares 5-10 Year Investment ...	Investment Grade ...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-07	2	0.1%	0.87
SPIB	State Street SPDR Portfolio I...	Investment Grade ...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-06	3	-0.1%	0.90
LQD	iShares iBoxx \$ Investment Gr...	Investment Grade ...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.2%	0.85
AGGY	WisdomTree Yield Enhanced U.S...	Aggregate Fixed I...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.3%	0.84
VCIT	Vanguard Intermediate-Term Co...	Investment Grade ...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-07	2	0.1%	0.90
IUSB	iShares Core Universal USD Bo...	Aggregate Fixed I...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.2%	0.87
BSV	Vanguard Short-Term Bond ETF	Fixed Income	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-06	3	-0.1%	0.91
QLTA	iShares Aaa A Rated Corporate...	Investment Grade ...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.2%	0.90
FBND	Fidelity Total Bond ETF	Active Fixed Inco...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.3%	0.87
VMBS	Vanguard Mortgage-Backed Secu...	Mortgage-Backed F...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.86
BOND	PIMCO Active Bond Exchange-Tr...	Active Fixed Inco...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.91
MBB	iShares MBS ETF	Mortgage-Backed F...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.87
GVI	iShares Intermediate Governme...	Fixed Income	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.93
AGG	iShares Core U.S. Aggregate B...	Aggregate Fixed I...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.89
BND	Vanguard Total Bond Market ETF	Aggregate Fixed I...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.90
SPAB	State Street SPDR Portfolio A...	Aggregate Fixed I...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.89
SCHZ	Schwab US Aggregate Bond ETF	Aggregate Fixed I...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.93
SPMB	State Street SPDR Portfolio M...	Mortgage-Backed F...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.89
AGZ	iShares Agency Bond ETF	Agency Fixed Inco...	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.98
SIVR	abrdn Physical Silver Shares ...	Silver	cluster_15	Mixed: DBP, GLTR, GDXJ	FLAT	-0.193	2026-05-11	0	0.0%	0.15
SLV	iShares Silver Trust	Silver	cluster_15	Mixed: DBP, GLTR, GDXJ	FLAT	-0.193	2026-05-11	0	0.0%	0.15
PPLT	abrdn Physical Platinum Share...	Platinum	cluster_15	Mixed: DBP, GLTR, GDXJ	FLAT	-0.193	2026-05-11	0	0.0%	0.50
SILJ	Amplify Junior Silver Miners ...	Materials	cluster_15	Mixed: DBP, GLTR, GDXJ	FLAT	-0.193	2026-05-11	0	0.0%	0.52
SIL	Global X Silver Miners ETF	Materials	cluster_15	Mixed: DBP, GLTR, GDXJ	FLAT	-0.193	2026-05-11	0	0.0%	0.59
XLP	State Street Consumer Staples...	Consumer Staples	cluster_17	Consumer Staples (6/1...	LONG	-0.232	2026-05-05	4	-0.8%	0.00

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KXI	iShares Global Consumer Stapl...	Consumer Staples	cluster_17	Consumer Staples (6/1...	LONG	-0.232	2026-05-06	3	-1.6%	0.02
IYK	iShares U.S. Consumer Staples...	Consumer Staples	cluster_17	Consumer Staples (6/1...	LONG	-0.232	2026-05-06	3	-0.2%	0.01
VDC	Vanguard Consumer Staples ETF	Consumer Staples	cluster_17	Consumer Staples (6/1...	LONG	-0.232	2026-05-01	6	-1.7%	0.04
FSTA	Fidelity MSCI Consumer Staple...	Consumer Staples	cluster_17	Consumer Staples (6/1...	LONG	-0.232	2026-05-01	6	-1.8%	0.04
PSCC	Invesco S&P SmallCap Consumer...	U.S. Small Cap Co...	cluster_17	Consumer Staples (6/1...	LONG	-0.232	2026-04-28	9	-2.2%	0.77
UUP	Invesco DB USD Index Bullish ...	Currency	cluster_26	Currency (1/2)	FLAT	-1.000	2026-04-27	10	-0.4%	0.47

Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

ticker	name	category	cluster_id	cluster_size	etf_signal	max_corr_to_others	median_corr_to_others	abs_pc1_loading	compo
RSX			cluster_18	11	LONG	0.371	0.052	0.00021	
UNG	United States Natural Gas Fund ...	Natural Gas	cluster_18	11	FLAT	0.492	0.081	0.00017	
GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short Fi...	cluster_23	9	FLAT	0.692	0.040	0.00007	
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short Fi...	cluster_23	9	LONG	0.635	0.221	0.00034	
FXV	Invesco CurrencyShares Japanese...	Currency	cluster_24	6	FLAT	0.732	0.094	0.00022	
DBA	Invesco DB Agriculture Fund	Agricultural Commodit...	cluster_18	11	FLAT	0.591	0.174	0.00038	
FTSM	First Trust Enhanced Short Matu...	Cash / Ultra-Short Fi...	cluster_23	9	FLAT	0.787	0.130	0.00030	
BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short Fi...	cluster_23	9	FLAT	0.825	0.021	0.00000	
SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short Fi...	cluster_23	9	FLAT	0.825	0.089	0.00022	
NEAR	iShares Short Duration Bond Act...	Cash / Ultra-Short Fi...	cluster_23	9	FLAT	0.720	0.182	0.00041	
MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short Fi...	cluster_23	9	FLAT	0.656	0.188	0.00047	
ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short Fi...	cluster_23	9	FLAT	0.793	0.162	0.00038	
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_18	11	LONG	0.487	0.219	0.00056	
GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	cluster_23	9	LONG	0.793	0.179	0.00043	
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_5	58	LONG	0.844	0.124	0.00033	
FXF	Invesco CurrencyShares Swiss Fr...	Currency	cluster_24	6	LONG	0.818	0.203	0.00047	
CMBS	iShares CMBS Bond ETF	Mortgage-Backed Fixed...	cluster_5	58	LONG	0.876	0.160	0.00030	
LMBS	First Trust Low Duration Opport...	Mortgage-Backed Fixed...	cluster_5	58	FLAT	0.836	0.231	0.00044	
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_16	13	FLAT	0.877	0.250	0.00044	
VNM	VanEck Vietnam ETF	Vietnam Equity	cluster_18	11	FLAT	0.452	0.334	0.00082	
PPLT	abrdn Physical Platinum Shares ...	Platinum	cluster_15	13	LONG	0.750	0.269	0.00072	
SUB	iShares Short-Term National Mun...	Municipal Bond	cluster_16	13	LONG	0.877	0.263	0.00051	
KURE	KraneShares MSCI All China Heal...	China Equity	cluster_13	15	FLAT	0.804	0.271	0.00070	
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_12	17	LONG	0.751	0.357	0.00082	
EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_18	11	FLAT	0.514	0.374	0.00105	
EPHE	iShares MSCI Philippines ETF	Philippines Equity	cluster_18	11	FLAT	0.578	0.409	0.00105	
KBA	KraneShares Bosera MSCI China A...	China Equity	cluster_13	15	LONG	0.855	0.254	0.00070	
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	cluster_22	9	FLAT	0.864	0.332	0.00070	
ECH	iShares MSCI Chile ETF	Chile Equity	cluster_18	11	LONG	0.637	0.429	0.00116	
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_13	15	LONG	0.905	0.242	0.00062	

Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

	scope	parameter	value
short_ma	screen	short_ma	5
buy_ma	screen	buy_ma	50
sell_ma	screen	sell_ma	20
vol_window	screen	vol_window	20
horizon	screen	horizon	5
cluster_smoothing_window	screen	cluster_smoothing_window	3
sell_cushion_k	screen	sell_cushion_k	0.5
sell_cushion_sd_window	screen	sell_cushion_sd_window	20